

# Spatial integration of sensory input and motor output in *Pseudomonas aeruginosa* chemotaxis through colocalized distribution

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## eLife Assessment

This **valuable** study by Wu et al presents data on bacterial cell organization, demonstrating that the two structures that account for bacterial motility - the chemotaxis complex and the flagella - colocalize to the same pole in *Pseudomonas aeruginosa* cells. The work provides **convincing** results for the regulation underlying this spatial organization and its functioning.

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## Abstract

The opportunistic pathogen *Pseudomonas aeruginosa* serves as a model organism for studying multiple signal transduction pathways. The chemoreceptor cluster, a core component of the chemotaxis pathway, is assembled from hundreds of proteins. The unipolar distribution of receptor clusters has long been recognized, yet the precise mechanism governing their assembly remains elusive. Here, we directly observed the relative positions of the flagellar motor and chemoreceptor cluster using flagellar filament labeling and gene editing techniques. Surprisingly, we found that both are located at the same cell pole, with the distribution pattern controlled by the polar anchor protein FlhF. Additionally, the efficient assembly of the chemoreceptor cluster is partially dependent on the integrity of the motor structure. Furthermore, we discovered that overexpression of the chemotaxis regulatory protein CheY leads to high intracellular levels of the second messenger c-di-GMP, triggering cell aggregation. Therefore, the colocalization of the chemoreceptor cluster and flagellum in *P. aeruginosa* serves to avoid cross-pathway signaling interference, enabling cells to conduct various physiological activities in an orderly manner.

## Introduction

In biological systems, the process by which proteins self-assemble into organized complex structures is widespread<sup>1,2</sup>. Pole-to-pole protein oscillations in the *Min* system ensure that the FtsZ ring, a crucial component of cell division, is placed precisely in the middle of the cell body<sup>3,4</sup>. In response to antibiotic treatment and heat stress, some bacteria generate multiple protein foci throughout the cytoplasm<sup>5,6</sup>. The polar localization of the secretion system, such as the type VI secretion system, is mediated by targeting proteins and potentially facilitates host-pathogen interactions<sup>7,8</sup>. Thus, spontaneous assembly is accompanied by the formation of spatial patterns that organize the intracellular environment.

To grow and colonize in various ecological niches, bacteria have evolved a mechanism for migrating towards more favorable environments. The locomotion organ (the flagellar motor) and the chemotaxis pathway play crucial roles in achieving this goal. The former provides physical drive, while the latter offers directional guidance. The initiation of cell reorientation is controlled by motor switching, which is governed by the chemotaxis two-component signaling pathway. This pathway comprises transmembrane chemoreceptors, a histidine kinase CheA, an adaptor protein CheW, a response regulator CheY, and two adaptation proteins, CheB and CheR. Receptors collaborate with CheA and CheW to form chemotaxis complex in a hexagonal pattern<sup>9,10</sup>. Upon phosphorylation of CheY by the kinase at the complex, it freely diffuses to the motor and binds to FlhM, a component of the motor switch complex, to affect motor switching<sup>11–13</sup>. It is worth noting that there exists diversity among different bacterial species in the placement of flagellar motors<sup>14</sup>, whereas chemoreceptor clusters are typically found at one or both ends of the cell<sup>15</sup>. These two structural units are functionally connected. However, it remains unclear whether there is an interaction in their spatial distribution and what the specific regulatory mechanism might be.

*Pseudomonas aeruginosa*, a common human opportunistic pathogen, possesses four chemosensory pathways that perform distinct functions and are stimulated by signal binding to 26 chemoreceptors<sup>16</sup>. Among them, the proteins encoded by chemotaxis-related genes collectively constitute the F6 pathway. Previous studies have suggested that the receptors involved in this pathway localize to the old cell pole<sup>17</sup>, similar to the flagellum<sup>18</sup>, implying colocalization of the receptors and the flagellum at the same pole; however, direct evidence for this colocalization is still lacking. Furthermore, the polar anchor protein FlhF has been reported to guide the flagellum to grow at the cell pole<sup>19,20</sup>. However, the mechanism by which chemotaxis complex-related proteins aggregate and self-assemble at the cell pole is controversial. The distribution of the chemotaxis complex in the peritrichous bacterium *Escherichia coli* has been extensively studied and attributed to several mechanisms, including stochastic self-assembly<sup>21,22</sup>, membrane curvature sorting<sup>23,24</sup>, and inefficient clustering in the lateral region<sup>25</sup>. However, these mechanisms cannot explain the unipolar distribution pattern observed in *P. aeruginosa*. Unlike *E. coli*, specific genes responsible for the placement of the chemotaxis complex have been identified in several bacterial species. In *Caulobacter crescentus*, chemotaxis complex assemble at the new cell pole with the help of TipN and TipF proteins<sup>26</sup>. A tripartite ParC-ParP-CheA interaction network was reported to promote polar localization of chemotaxis complex in *Vibrio parahaemolyticus*<sup>27,28</sup>. However, as there are no related genes in *P. aeruginosa*, the mechanism of its chemotaxis complex distribution remains a mystery.

Here, we combined gene editing and in vivo fluorescence imaging of flagellar filaments to directly observe the distribution of chemotaxis complex and flagellar motors, proposing a cooperative construction model of chemotaxis network and flagella during the entire division cycle of *P. aeruginosa*. The core focus of this study is to clarify the regulatory mechanism of its chemotaxis complex distribution. We found a substantial association between the assembly of the flagellar

motor and the chemotaxis complex. The complete assembly of the motor serves as a partial prerequisite for the assembly of the chemotaxis complex, and its assembly site is also regulated by the polar anchor protein FlhF. Furthermore, by introducing exogenously expressed CheY protein, we found that this triggers the expression of c-di-GMP at a high level. From this, we infer that the colocalization of the chemotaxis complex and the flagellum in *P. aeruginosa* avoids the cross-pathway interference of signaling molecules, thus providing a guarantee for the coexistence of multiple chemosensory pathways.

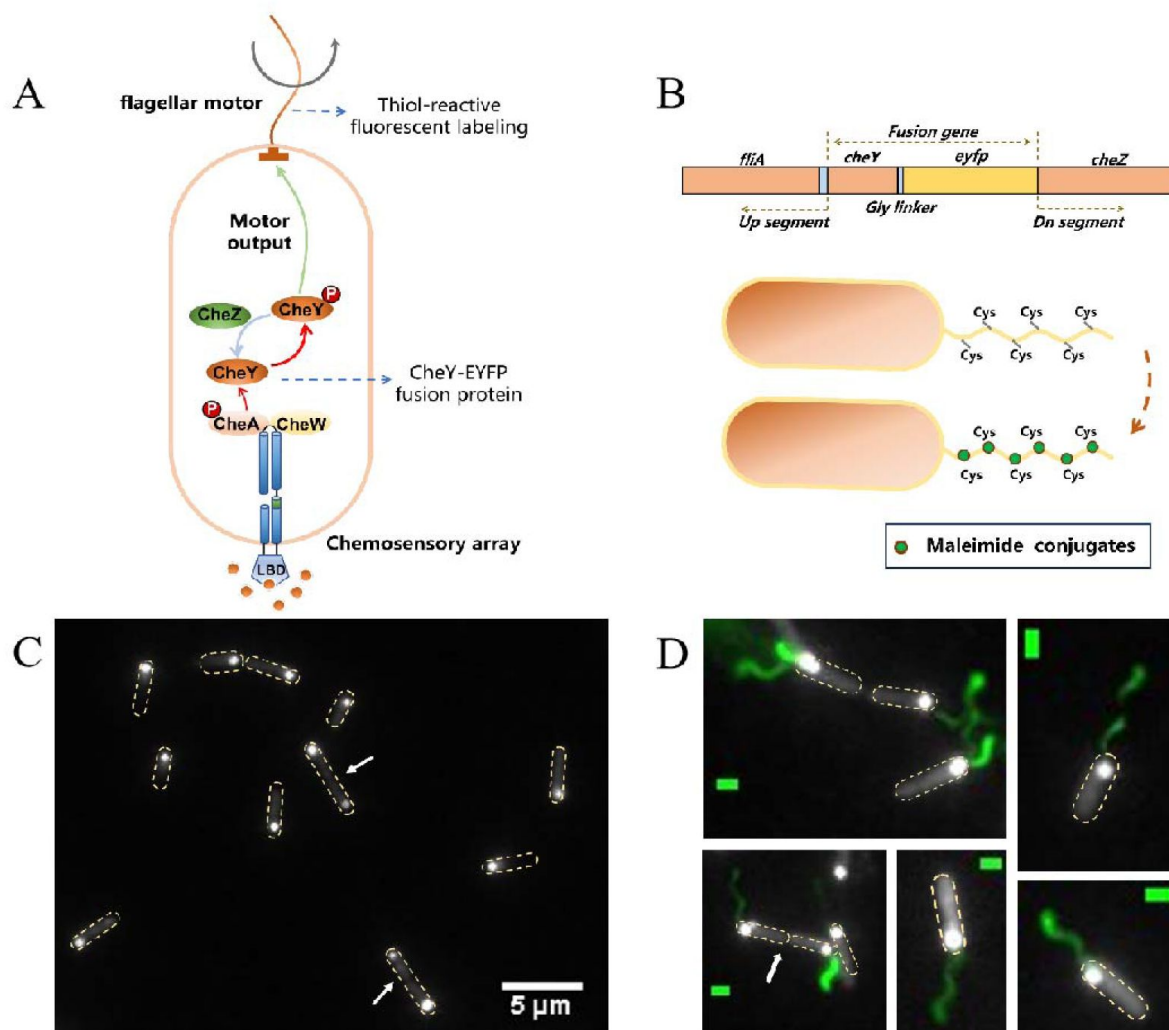
## Results

### Robust generation of daughter cell with both chemotaxis network and flagellar motor

CheY has been shown to colocalize with chemoreceptors<sup>17,29</sup>. To visualize the distribution of chemotaxis complex in cells, we fused the gene encoding the enhanced yellow fluorescent protein (eYFP) to the *cheY* gene in the *P. aeruginosa* chromosome (Fig. 1A). All genetic modifications were carried out at the native position on the chromosome, allowing the expression of chemotaxis proteins in precise stoichiometry from their natural promoters (Fig. 1B). The mutant strain with *cheY-eyfp* was able to moderately expand on soft agar plates<sup>17</sup>, proving that its chemotaxis behavior was not notably affected, so it was selected for subsequent experiments. As shown in Fig. 1C, CheY-eYFP was mainly located at the cell pole, suggesting that CheA, CheW and CheY in *P. aeruginosa* form a signal transduction complex that was primarily distributed at the cell pole, as described previously<sup>17</sup>. Representative large-field images containing many cells are shown in Fig. S1. Additionally, we constructed a plasmid expressing CheA-CFP and electroporated it into the *cheY-eyfp* strain. Fluorescence imaging revealed a high degree of spatial overlap between CheA-CFP and CheY-EYFP (Fig. S2), confirming that CheY-EYFP accurately marks the location of the chemoreceptor complex. From our measurements, nearly 90% (332/372) of cells contain obvious receptor clusters, and factors such as fluorescence bleaching may have caused the loss of fluorescent spots in some cells. In addition, we observed the presence of fluorescent spots at both ends of some cells with a large aspect ratio (probably approaching cell division), indicating that the newly generated progeny cells will have a complete chemotaxis network. Next, we sought to observe the distribution of the flagella in cells simultaneously to understand the motility of future progeny.

Our recent study demonstrated the successful labeling of the flagellar filament with thiol-reactive fluorescent dye by introducing cysteines into the flagellin FliC<sup>30,31</sup> (Fig. 1B). To avoid fluorescence interference, we utilized a dye with an excitation peak near 568 nm, considering that the excitation peak of eYFP is 513 nm, and the emission peak is 527 nm. We employed a xenon lamp as the excitation light source and switched filters to enable simultaneous fluorescence imaging of flagellar filaments and chemotaxis complex within the same cell.

Remarkably, our findings revealed a surprising colocalization of chemotaxis complex and flagellar filaments at the same end of the cell body, and this colocalization was consistent, as shown in Fig. 1D. A typical large field was shown in Fig. S3. Similar to the chemotaxis complex, we observed the phenomenon of double flagella symbiosis in cells with a large aspect ratio, suggesting that *P. aeruginosa* has evolved a cell division mechanism with precise timing regulation. Before cell division, both poles of the mother cell assembled chemotaxis complex and flagellar motors. This ensures that daughter cells possess complete motility and chemotaxis, thereby greatly enhancing the environmental adaptability of the population.



**Fig. 1.**

**A.** Schematic diagram of the chemotaxis signal transduction network and flagellar motor of *P. aeruginosa*. Fusion protein CheY-EYFP and fluorescently labeled flagellar filaments were used as markers to indicate the position of the chemotaxis complex and motor, respectively. **B.** The labeling mechanism of flagellar filaments and chemotaxis regulatory protein CheY. Filaments (with cysteine point mutation FliC<sup>T394C</sup>) were labeled through sulfhydryl-maleimide conjugation, and *cheY-eyfp* fusion with a 3x glycine linker was used to visualize chemotaxis complex positions. **C.** Localization of CheY-EYFP in the wild-type strain of *P. aeruginosa*. CheY-EYFP is mainly located at the single cell pole, and the white arrow points to individuals with obvious chemotaxis complex at both cell poles, which generally have a large aspect ratio of the cell body. The yellow dashed box marks the cell outline. **D.** The merged imaging of flagellar filaments and CheY-EYFP in the wild-type strain of *P. aeruginosa*, where flagellar motor and chemotaxis complex colocalize in cells. 145 cells with labeled flagella were observed, all of which exhibited consistent colocalization. White arrows point to individuals about to be divided, and the yellow dashed box marks the cell outline. The scale bar is 1  $\mu$ m.

## FlhF mediates proper unipolar localization of the chemotaxis complex

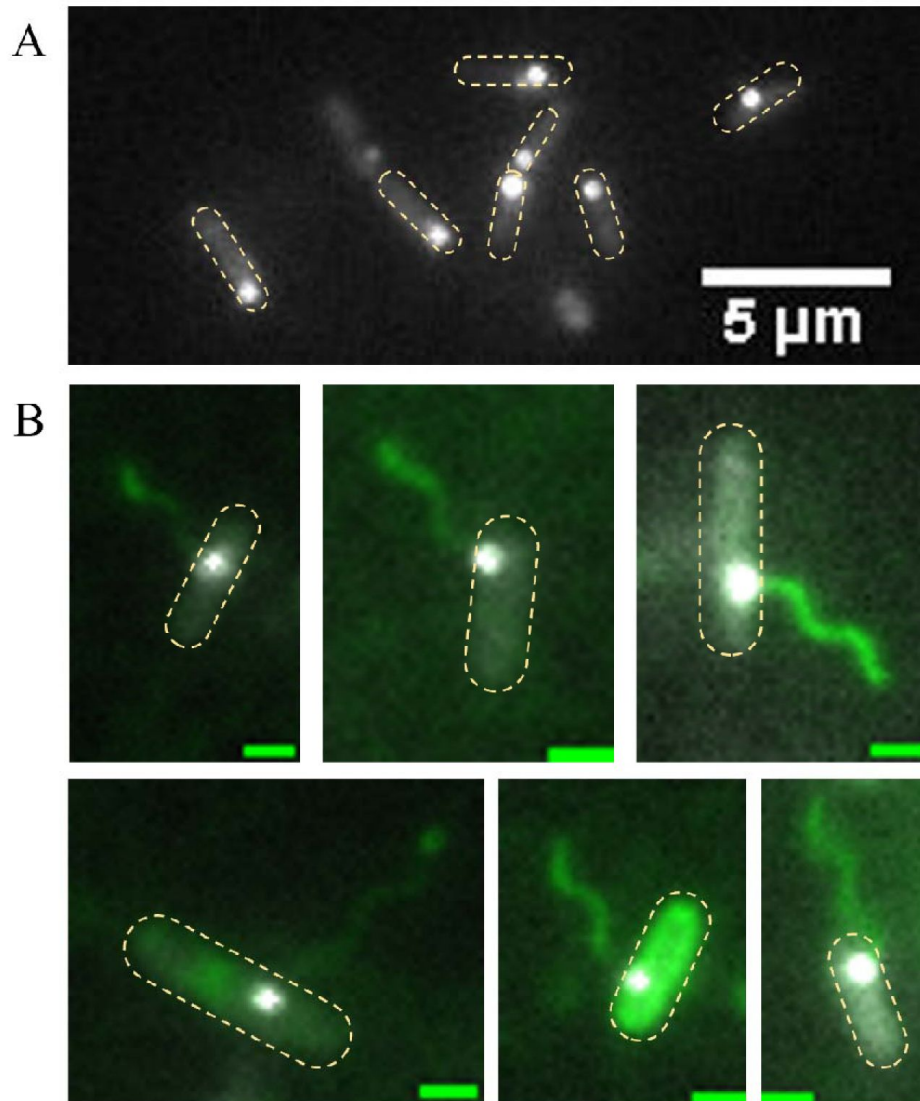
Despite potential differences in the physical and especially physiological environments at the two cell poles, it is unlikely that the unipolar distribution of the chemotaxis complex can be attributed to passive regulatory factors. The number and distribution of flagellar motors and chemotaxis complex vary among different bacterial species<sup>14,15</sup>, and the localization system for flagella has been well studied. Specifically, the FlhF-FlhG system, discovered in several monotrichous bacterial species, has been shown to control the location and number of flagella<sup>32,33</sup>. In *P. aeruginosa*, a knockout of *flhF* leads to mis-localized flagellar assembly<sup>19,20</sup>. Considering the consistent colocalization pattern between chemotaxis complex and flagellar motors in *P. aeruginosa*, we speculate that the distribution of chemotaxis complex is also regulated by similar molecular mechanisms.

To investigate the role of FlhF in the localization of chemotaxis complex, we constructed a  $\Delta flhF$  strain for fluorescence observation. The results revealed that the chemotaxis complex no longer grow robustly at the cell pole (**Fig. 2A**), and the assembly positions of the flagellar motor change accordingly, colocalizing with the chemotaxis complex (**Fig. 2B**). Representative large fields containing many cells are shown in **Fig. S1**. Additionally, we categorized receptor cluster distribution patterns into three types: precise polar, near polar, and mid-cell. The relevant statistics for the wild-type strain and the  $\Delta flhF$  mutant are presented in **Fig. S4**, which show a significant increase in the proportion of non-precise polar chemoreceptors following *flhF* knockout. We also quantified the proportion of individuals with obvious receptor clusters, which was more than 80% (182/221), similar to the wild-type strain. These experimental results suggest that the polar anchoring protein FlhF has a minimal impact on the assembly efficiency of *P. aeruginosa*'s chemotaxis complex, but it strictly controls its unipolar distribution. Furthermore, a consistent colocalization between the flagellar motor and chemotaxis complex was observed, independent of FlhF. Thus, it is crucial to determine whether there is a causal relationship in the assembly order of these two structural units.

## Motor structural integrity is a prerequisite for chemoreceptor self-assembly

Cryo-electron microscopy has successfully revealed the complete flagella structure in various bacterial species. This structure includes multiple components such as the inner-membrane MS ring, the cytoplasmic C ring, and the internal protein secretion system, all of which exhibit strong similarities<sup>34</sup>. This suggests that the core structure of the flagellar motor is highly conserved. The MS ring, composed of 26 FliF proteins, forms the foundation of the flagellar structure<sup>35</sup>. In addition to serving as a mounting platform for the C ring, the MS ring also acts as a protective shell for the internal protein secretion machinery<sup>36</sup>. FliG, the main component protein of the C ring, binds directly to the cytoplasmic surface of FliF in a 1:1 ratio<sup>37</sup>. Thus, the efficient assembly of the C ring requires preassembly of the MS ring. To observe the distribution of the chemotaxis complex following the disruption of the C ring and MS ring, we constructed  $\Delta fliG$  and  $\Delta fliF$  mutants.

We performed fluorescence observation on the  $\Delta fliG$  mutant, which has a disrupted flagellar motor C ring. The results showed that the proportion of  $\Delta fliG$  cells with obvious chemoreceptor clusters decreased significantly compared to the wild-type cells (59.8%, 140/234), although the chemotaxis complex remained at a single cell pole. We also examined the  $\Delta fliF$  mutant, which has a disrupted flagellar motor MS ring. Similar to the  $\Delta fliG$  mutant, the proportion of individuals with obvious chemotaxis complex decreased significantly (62.5%, 202/323). Subsequently, we constructed a  $\Delta flhF\Delta fliF$  mutant and observed that the proportion of individuals with obvious



**Fig. 2.**

**A.** Localization of CheY-EYFP in the  $\Delta flhF$  strain of *P. aeruginosa*. CheY-EYFP is no longer robustly distributed at the single cell pole. The yellow dashed box marks the cell outline. **B.** The merged imaging of flagellar filaments and CheY-EYFP in the  $\Delta flhF$  strain of *P. aeruginosa*, flagellar motor and chemotaxis complex still colocalize in cells. 101 cells with labeled flagella were observed, all of which exhibited consistent colocalization. The yellow dashed box marks the cell outline. The scale bar is 1  $\mu$ m.

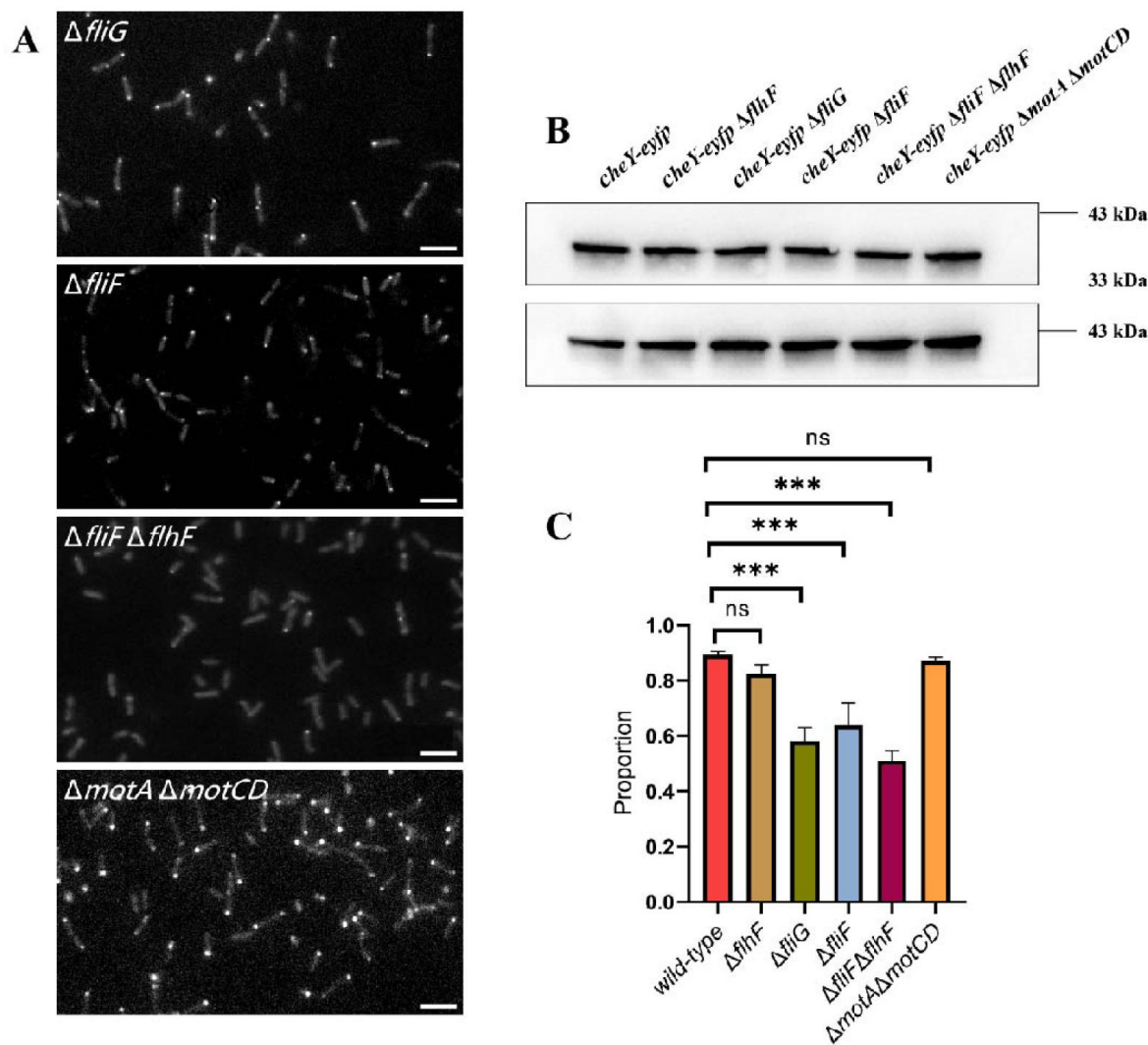


chemoreceptor clusters further decreased (50.7%, 341/672). To ascertain whether it is motor integrity rather than functionality that influences the efficiency of chemotaxis complex assembly, we constructed a stator mutant ( $\Delta\text{motA}\Delta\text{motCD}$ ). In this mutant, the motor is completely stalled while the structure remains intact. We found that the mutant performed similarly to the wild-type strain in terms of chemotaxis complex assembly (84.3% of individuals with obvious clusters, 204/242). The fluorescence imaging of receptor clusters for multiple mutants in this section is shown in **Figure 3A**, and corresponding large-field images are provided in **Fig. S1**. We utilized Western blotting to measure the expression levels of CheY, which were found to be similar across the different strains (**Fig. 3B**). These further substantiated that the observed phenomenon is based on the structural integrity of the motor rather than the protein expression level. In contrast, the *cheA* knockout strain was constructed to disrupt the assembly of the chemoreceptor complex. We fluorescently labeled its flagellar filaments and found that its phenotype was no different from that of the wild-type strain (**Fig. S5**), indicating that the assembly of the chemotactic receptor complex does not affect the flagellar assembly efficiency. The P ring (mainly composed of FlgI) is thought to act as a bushing for the peptidoglycan layer, and its absence results in partial assembly of the motor structure<sup>38</sup>. We constructed  $\Delta\text{flgI}$  mutant and found that the proportion of cells with distinct chemotactic complexes was similar to that of the wild-type (**Fig. S6**). Additionally, we quantified the proportion of cells with receptor clusters (**Fig. 3C**). Overall, our findings suggest that the polar anchor protein FlhF and the structural integrity of the motor are crucial for the formation of chemotaxis complex. The former guides them to the appropriate site, while the latter influences the assembly efficiency of the chemoreceptor complex.

## Colocalization of chemotaxis complex and flagellar motor avoids cross-pathway interference in *P. aeruginosa* signal transduction

The distribution patterns of receptors associated with various signal transduction pathways in *P. aeruginosa* vary considerably. For instance, the biofilm formation-related receptor WspA, unlike the chemotaxis receptors discussed here, is distributed throughout the cell. This distribution is thought to enhance its sensitivity to mechanical perturbations of the cell membrane<sup>39</sup>. Previous studies speculated that the highly consistent position of chemotaxis complex and flagellar motors enhances bacterial chemotaxis performance<sup>28</sup>. However, the diffusion time of the phosphorylated chemotaxis regulatory protein CheY-P across the longest distance in a bacterial cell body (along the cell's long axis) is approximately 100 ms<sup>40,41</sup>, whereas the timescale for the chemotaxis temporal comparison is on the order of seconds<sup>42</sup>. Additionally, a study by Fukuoka and colleagues reported that intracellular chemotaxis signal transduction requires approximately 240 ms beyond CheY or CheY-P diffusion time<sup>41</sup>. Moreover, the CCW/CW interval of the *P. aeruginosa* flagellar motor under normal conditions is 1-2 s, as determined by bead assays<sup>30</sup> or tethered cell assays<sup>43</sup>. Taken together, these indicate that for *P. aeruginosa*, which moves via a run-reverse mode, the potential 100 ms reduction in response time due to co-localization of the chemotaxis complex and motor has a limited effect on overall chemotaxis timing. Consequently, the physiological significance of the colocalization of chemoreceptors and flagellar motors remains unclear.

*E. coli* mediates chemotaxis through a single pathway involving five types of chemoreceptors<sup>44</sup>. As a core regulatory protein for chemotaxis, CheY participates in multiple processes such as adaptation and chemotaxis response, with CheY-P binds to the motor to regulate switching<sup>11</sup>. In contrast, *P. aeruginosa* has four distinct chemosensory pathways that perform different functions, involving different CheY homologs, and are stimulated by signals binding to 26 types of chemoreceptors. Consequently, *P. aeruginosa* possesses more complex chemosensory pathways<sup>16</sup>. We thus hypothesized that the co-localization of chemoreceptors and flagellar



**Fig. 3.**

**A.** Localization of CheY-EYFP in various *P. aeruginosa* strains. The scale bar is 10  $\mu m$ . **B.** Western blot analysis was performed to detect CheY expression in various *P. aeruginosa* strains.  $\beta$ -actin was used as the housekeeping protein. **C.** Occurrence probability of obvious chemotaxis complex in *P. aeruginosa* wild-type and several mutant strains. The proportion of individuals with obvious chemotaxis complex decreased significantly in the motor-incomplete strains ( $\Delta fliF$  and  $\Delta fliG$ ), and this value was further reduced after *flhF* knockout ( $\Delta fliF \Delta flhF$ ). The  $\Delta fliF$  and  $\Delta motA \Delta motCD$  strains have a similar chemotaxis complex occurrence probability as the wild-type strain. The number of cells analyzed for each strain (from left to right) were: 372, 221, 234, 323, 672, and 242. '\*\*\*': significant difference (P-value<0.0001), 'ns.': no significant difference (P-value>0.05).



motors in *P. aeruginosa* ensures locally distributed CheY-P molecules, thereby eliminating the need for a high level of intracellular CheY-P and avoiding potential side effects on other signaling pathways.

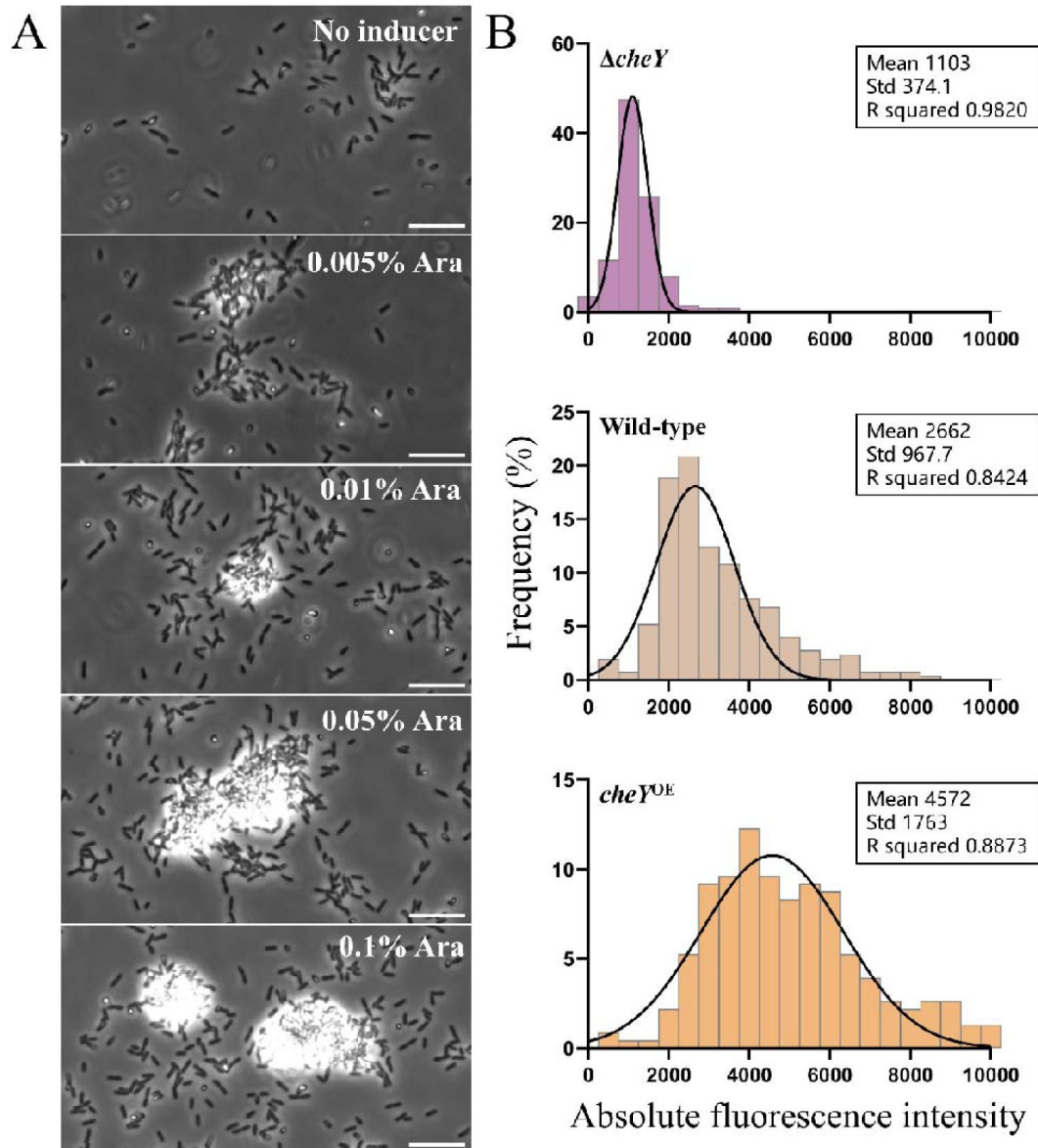
To test the potential effect of an increased intracellular CheY-P level, we constructed the CheY expression plasmid *cheY*-pJN105, transformed it into the wild-type *P. aeruginosa* strain, and induced it with varying concentrations of arabinose. We observed that the higher the inducer concentration, the more pronounced the cell aggregation became in the field of view (**Fig. 4A**). The transition from planktonic individuals to aggregated communities is similar to biofilm formation, which is known to be accompanied by a significant increase in intracellular c-di-GMP levels.

To investigate whether a higher level of intracellular CheY-P increased the intracellular c-di-GMP level, we sought to detect the c-di-GMP level. We introduced the plasmid pCdrA-gfp into *P. aeruginosa*, using it as a c-di-GMP biosensor with the fluorescence intensity proportional to the c-di-GMP level<sup>45</sup>. To ensure the coexistence of the plasmids, the intracellular CheY concentration was controlled by the plasmid *cheY*-pME6032. The intracellular CheY levels induced by varying arabinose concentrations were measured, showing clear differences across the concentration gradient (**Fig. S7**). We found that the average single-cell fluorescence intensity of the CheY overexpression strain was 71.8% higher than that of the wild-type strain, while the c-di-GMP level of the *cheY* deletion strain was 58.6% lower than that of the wild-type (**Fig. 4B**).

The colocalization of both the signaling source (the kinase) and sink (the phosphatase) at the chemoreceptor complex at the cell pole results in a rapid decay of CheY-P concentration within approximately 0.2  $\mu\text{m}$  from the cell pole, leading to a nearly uniform distribution elsewhere in the cell, as demonstrated by Vaknin and Berg<sup>46</sup>. This spatial arrangement effectively confines high CheY-P levels to the pole region. When the motor is also localized at the cell pole, the need for elevated CheY-P concentrations throughout the cytoplasm is reduced. Therefore, the colocalization of chemotaxis complex and flagellar motor facilitates the precise regulation of cell swimming direction within the low intracellular CheY-P concentration threshold, using locally distributed CheY-P molecules. This helps to avoid the occurrence of the aforementioned cross-pathway interference. To further confirm that CheY overexpression promotes aggregation through increased c-di-GMP levels, we performed additional experiments co-overexpressing CheY and a phosphodiesterase (PDE) from *E. coli* to reduce intracellular c-di-GMP. These experiments showed that PDE expression mitigates cell aggregation caused by CheY overexpression in *P. aeruginosa* (**Fig. S8**).

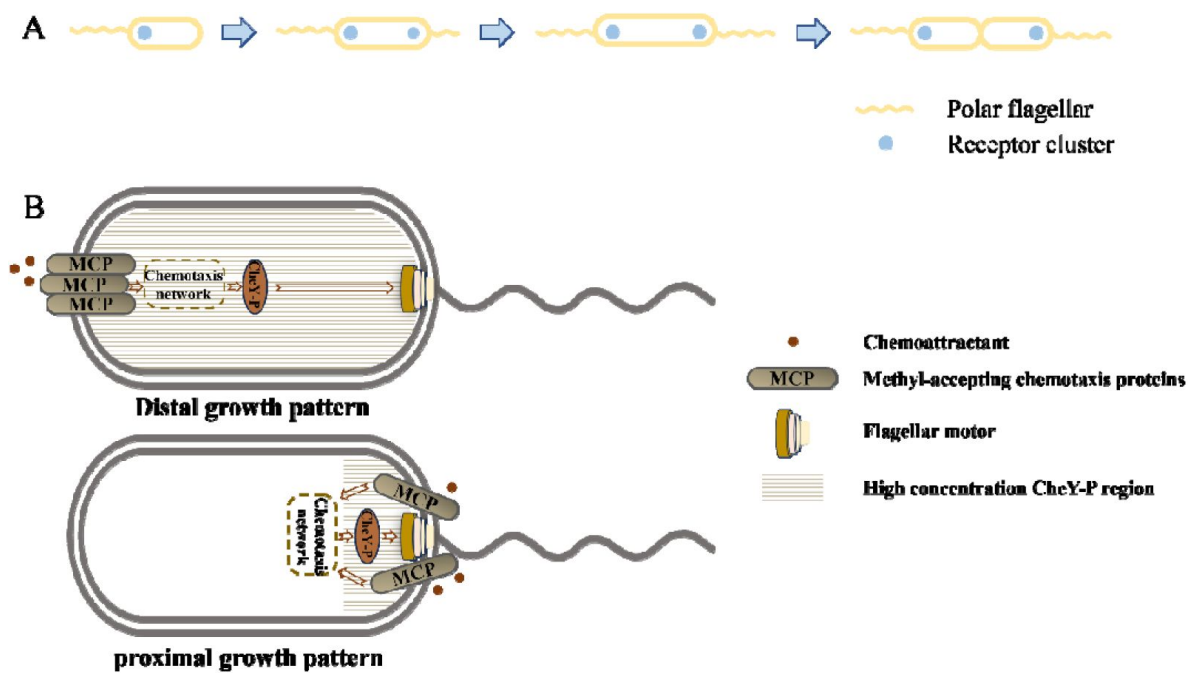
## Summary and discussion

*P. aeruginosa* harbors multiple signal transduction systems that regulate flagella-mediated swimming motility (Che pathway), pili-mediated interface twitching motility (Pil pathway), and biofilm formation (Wsp pathway)<sup>16</sup>. These systems allow it to thrive in various ecological niches within complex external environments. Here, employing a combination of chromosomal fluorescent protein fusion and flagellar labeling techniques in living cells, we directly observed that the chemotaxis complex of *P. aeruginosa* Che pathway and the flagellar motor share a high degree of consistency in their assembly sites. We found that the chemotaxis complex, comprising Che proteins, was consistently located at the cell pole where the flagellum was positioned. Based on these observations, we deduced the construction mode of the chemotactic network and flagellar motor during the entire cell growth cycle (**Fig. 5A**). As the cell body matures and elongates, a new flagellum grows at the opposing cell pole, accompanied by the assembly of a fresh chemotaxis complex. The cell then undergoes division from the middle, generating two daughter cells, each equipped with fully functional motility and chemotaxis capabilities. This ensures robust inheritance of these chemotaxis and motility-related macromolecular machines.



**Fig. 4.**

**A.** The evolution of cell aggregation as the intracellular CheY concentration increases by induction with higher concentrations of arabinose. The scale bar is 10  $\mu$ m. **B.** Quantitative characterization of intracellular c-di-GMP levels at different CheY concentrations. From top to bottom, they correspond to the  $\Delta cheY$  strain (N=198), wildtype strain (N=249) and CheY overexpression strain (N=228), respectively.



**Fig. 5.**

**A.** The construction mode of the chemotaxis network and flagellar motor throughout the complete cell growth cycle. **B.** The proximal growth mode of the *P. aeruginosa* flagellar motor and receptor clusters will effectively regulate the spatial range of CheY action, thus avoiding unintentional cross-pathway regulation.

Based on our understanding of the regulation mechanism of flagellar position in *P. aeruginosa*, we constructed a  $\Delta flhF$  mutant and found that the position distribution of its chemotaxis complex is actively regulated by FlhF-mediated molecular mechanisms, rather than being passively restricted by membrane curvature factors as seen in *E. coli*. A similar active regulation phenomenon was previously reported in *Vibrio cholerae*, with the corresponding molecular regulation mechanism detailed extensively<sup>28</sup>. The flagella and chemotaxis complex of newborn *V. cholerae* cells are located at the old cell pole, with the polar anchor protein HubP distributed at both cell poles. As the cell grows, HubP recruits the ParA homologue ParC. Subsequently, ParC recruits ParP to assemble a new chemotaxis complex at the new cell pole. Upon cell division, HubP relocates to the middle of the cell, ensuring its robust presences at both poles of the daughter cells. Here, HubP recruits ParA homologues including FlhG, which influence flagellar motor assembly. However, previous experiments confirmed that the cell cycle-dependent polar localization of ParC and chemotactic proteins is independent of the flagellar position regulatory protein FlhF. This suggests that the localization of flagellar motors and chemotaxis complex in *V. cholerae* is controlled separately through different pathways. In contrast, flagellar motors were no longer robustly distributed at the cell pole after *flhF* was knocked out in *P. aeruginosa*, whereas the assembly site of chemotaxis complex and flagellar motor remained remarkably consistent. This suggests that FlhF regulates the distribution patterns of both and implies an interaction between them.

We further constructed  $\Delta fliG$  and  $\Delta fliF$  mutants, which disrupted the assembly of the flagellar motor. Surprisingly, we observed that the assembly of chemotaxis complex was impaired under these conditions, with a significant decrease in the proportion of individuals exhibiting obvious Che protein fluorescent bright spots. In contrast, the chemotaxis complex assembly in the stator mutant ( $\Delta motA\Delta motCD$ ) was similar to that of the wild-type strain. It should be noted that *P. aeruginosa* has a four-tiered transcriptional regulatory circuit controlling flagellar biogenesis, in which FliG and FliF belong to class II and are regulated by the master transcriptional regulator FleQ<sup>47</sup>.

Chemotaxis complex-related genes (*cheA*, *cheW*, etc.) are regulated by intracellular free FliA proteins, and there is no evidence that FliA is regulated by proteins such as FliG. To further confirm that the expression of core chemotaxis proteins was not affected by the knockout of *fliG*, we performed Western blotting to measure CheY expression levels in wild-type,  $\Delta fliF$  and  $\Delta fliG$  strains, and found no significant differences (Fig. 3B). These results suggested that the integrity rather than functionality of the flagellar motor in *P. aeruginosa* is critical for the assembly of its chemotaxis complex, and its assembly efficiency is greatly reduced in the absence of flagellar motors. To eliminate the potential confounding effects of FlhF and flagellar motors, we constructed a  $\Delta flhF\Delta fliF$  mutant strain. Under this condition, the proportion of individuals with obvious Che protein fluorescent bright spots dropped to ~50%, indicating that passive regulation based on membrane curvature may still exist in *P. aeruginosa*, albeit to a lesser degree.

Is the remarkable consistency in the spatial distribution of these two independent structural units an unintentional outcome or a hidden mystery? Given that Kulasekara et al. previously proposed that CheA and its phosphorylation state can cause c-di-GMP heterogeneity among individual *P. aeruginosa* cells through a specific PDE<sup>48,49</sup>, we sought to verify whether the phenotypic differences induced by CheY here are closely related to changes in the overall population level of c-di-GMP. We introduced a c-di-GMP monitoring system at the single-cell level and discovered that overexpression CheY substantially increases the intracellular c-di-GMP concentration. Previous studies generally believed that CheY-P, as a chemotactic regulatory protein, only influences flagellar motor switching. Here, we found that CheY-P, as a signaling molecule, can accomplish threshold-limited cross-pathway regulation. At low CheY-P concentrations, it is conservatively involved in motor switching regulation, as confirmed by previous studies. At high CheY-P concentrations, cell motility is suppressed, and c-di-GMP is triggered to express, though its specific mechanism needs further exploration. Unlike CheA, a structural protein of the chemotaxis complex, CheY is a protein that can shuttle in the cytoplasm and can also regulate the intracellular

c-di-GMP level. This finding expands our understanding of the connection between the internal chemotaxis network and the c-di-GMP pathway. The intracellular environment is highly crowded<sup>50</sup>, and the proximal growth pattern shown in **Fig. 5B** has multiple physiological significances. On one hand, chemotaxis-related protein can precisely regulate cell motility with minimal synthesis costs. On the other hand, the chemotactic network and the flagellar motor are spatially integrated within the cell, ensuring their robust existence within the cell body and preventing mutual interference with collateral pathways.

Many bacteria possess multiple signal transduction pathways. The orderly operation of each pathway within the micrometer-scale space presents a fascinating scientific problem. Here, we utilized the well-known chemotaxis network as a starting point to shed light on this problem, providing inspiration for future in-depth understanding of related issues.

## Materials and methods

### Strains and cell culture

The strains and plasmids used in this study are listed in **Table 1**. The *Escherichia coli* TOP10 strain was used for standard genetic manipulations. A single-colony isolate was grown in 3 ml of LB broth (1% Bacto tryptone, 0.5% yeast extract and 1% NaCl) overnight to saturation on a rotary shaker (250 rpm) at 37°C. An aliquot was diluted 1:100 into 10 ml of LB broth and grew to exponential phase. Appropriate antibiotics were added if necessary to prevent plasmid loss: for *E. coli*, 15 µg/ml Gentamicin and 25 µg/ml Tetracycline; for *P. aeruginosa*, 30 µg/ml Gentamicin and 50 µg/ml Tetracycline. To induce protein expression, Isopropyl β-D-1-thiogalactopyranoside (IPTG, 1.0 mM) was added to strains with pME6032-derivative vectors, and arabinose (0.005%, 0.01%, 0.05%, 0.1%) was added to strains with pJN105-derivative vectors. 2 ml of cells were harvested by centrifugation at 2000×g for 2 min, washed twice in an equal volume of motility buffer (MB) [50 mM potassium phosphate, 15 µM EDTA, 0.15 M NaCl, 5 mM Mg<sup>2+</sup> and 10 mM lactic acid (pH 7.0)]<sup>51</sup>, and resuspended in 5 ml MB for subsequent fluorescence observation.

### Construction of in-frame deletion mutants

We used polymerase chain reaction (PCR) to generate ~1000 bp DNA fragments with upstream (Up) and downstream (Dn) sequences flanking the target gene (including *flhF*, *fliF*, *motA* and *fliG*) to be deleted. The Up and Dn DNA sequences were fused with the linearized pex18gm vector using Gibson assembly<sup>52</sup>. The resulting vectors were transformed into *P. aeruginosa* by electroporation, and the desired knockout mutants were obtained by double selections on gentamicin plates (LB plates with 30 µg/ml gentamicin) and sucrose plates (NaCl-free LB plates with 15% sucrose) at 37°C<sup>53</sup>.

### Generation of chromosomal fusions of *yfp* to *P. aeruginosa* gene

To generate *eyfp* fusion at the N-terminus of *cheY*, we first used PCR to generate upstream and downstream DNA sequences flanking the *cheY* gene. A subsequent PCR reaction aimed to yield the *cheY-eyfp* fusion with a 3× glycine linker (GGCGGAGGA), with pVS88 serving as the source of the enhanced yellow fluorescent (*eyfp*). The three DNA sequences, along with the linearized vector, were fused to *cheY-eyfp*-pex18gm using Gibson assembly. Finally, gene replacement was used to transfer the fusion constructs into the *P. aeruginosa* chromosome by homologous recombination.

### Flagellar staining and fluorescence imaging

Flagellar filaments were labeled by following the protocol described previously<sup>54</sup>. Cells (1 ml of exponential-phase culture) were harvested by centrifugation at 2000×g for 10 min and washed twice in 1 ml MB. The final pellet was adjusted to a volume of ~100 µl that concentrated the



| Strain, Plasmid                                                                               | Genotype, phenotype, and Description                                                                                                                                                                                                                                    | Source        |
|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Strains</b>                                                                                |                                                                                                                                                                                                                                                                         |               |
| <i>P. aeruginosa</i>                                                                          |                                                                                                                                                                                                                                                                         |               |
| PAO1                                                                                          | wild-type strain                                                                                                                                                                                                                                                        | Fan Jin Group |
| PAO1 <i>fliC</i> <sup>T394C</sup>                                                             | Replacement of chromosomal <i>fliC</i> in PAO1                                                                                                                                                                                                                          | Ref 31        |
| PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i>                                            | <i>yfp</i> fusions at the N-terminus of <i>cheY</i> in PAO1 <i>fliC</i> <sup>T394C</sup>                                                                                                                                                                                | This work     |
| PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i> $\Delta$ <i>flhF</i>                       | Nonpolar <i>flhF</i> deletion in PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i>                                                                                                                                                                                     | This work     |
| PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i> $\Delta$ <i>fliF</i>                       | Nonpolar <i>fliF</i> deletion in PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i>                                                                                                                                                                                     | This work     |
| PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i> $\Delta$ <i>fliG</i>                       | Nonpolar <i>fliG</i> deletion in PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i>                                                                                                                                                                                     | This work     |
| PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i> $\Delta$ <i>flhF</i> $\Delta$ <i>fliF</i>  | Nonpolar <i>fliF</i> deletion in PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i> $\Delta$ <i>flhF</i>                                                                                                                                                                | This work     |
| PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i> $\Delta$ <i>motA</i> $\Delta$ <i>motCD</i> | Nonpolar <i>motAB</i> and <i>motCD</i> deletion in PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i>                                                                                                                                                                   | This work     |
| PAO1 <i>fliC</i> <sup>T394C</sup> $\Delta$ <i>cheA</i>                                        | Nonpolar <i>cheA</i> deletion in PAO1 <i>fliC</i> <sup>T394C</sup>                                                                                                                                                                                                      | This work     |
| PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i> $\Delta$ <i>flgI</i>                       | Nonpolar <i>flgI</i> deletion in PAO1 <i>fliC</i> <sup>T394C</sup> <i>cheY-eyfp</i>                                                                                                                                                                                     | This work     |
| <i>E. coli</i>                                                                                |                                                                                                                                                                                                                                                                         |               |
| Top10                                                                                         | <i>F-mcrA</i> $\Delta$ ( <i>mrr-hsRMS-mcrBC</i> ) $\Phi$ 80 <i>lacZ</i> $\Delta$ <i>M15</i> $\Delta$ <i>lacX74</i> <i>recA1</i> <i>araD139</i> $\Delta$ ( <i>araIeu</i> ) 7697 <i>galU</i> <i>galK</i> <i>rpsL</i> ( <i>Nal</i> <sup>R</sup> ) <i>endA1</i> <i>mupG</i> | Invitrogen    |
| <b>Plasmids</b>                                                                               |                                                                                                                                                                                                                                                                         |               |
| pex18gm                                                                                       | oriT <sup>+</sup> <i>sacB</i> <sup>+</sup> ; gene replacement vector with MCS from pUC18; Gm <sup>r</sup>                                                                                                                                                               | Fan Jin Group |
| <i>flhF</i> -pex18gm                                                                          | In-frame deletion of <i>flhF</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                   | This work     |
| <i>fliF</i> -pex18gm                                                                          | In-frame deletion of <i>fliF</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                   | This work     |
| <i>fliG</i> -pex18gm                                                                          | In-frame deletion of <i>fliG</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                   | This work     |
| <i>motA</i> -pex18gm                                                                          | In-frame deletion of <i>motA</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                   | This work     |
| <i>motCD</i> -pex18gm                                                                         | In-frame deletion of <i>motCD</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                  | Ref 30        |
| <i>cheA</i> -pex18gm                                                                          | In-frame deletion of <i>cheA</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                   | This work     |
| <i>flgI</i> -pex18gm                                                                          | In-frame deletion of <i>flgI</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                   | This work     |
| <i>cheY-eyfp</i> -pex18gm                                                                     | <i>eyfp</i> fusions <i>cheY</i> cloned into pex18gm; Gm <sup>r</sup>                                                                                                                                                                                                    | This work     |
| <i>cheY</i> -pJN105                                                                           | <i>cheY</i> overexpression vector in pJN105, <i>cheY</i> expression is controlled by P <sub>BAD</sub> promoter; Gm <sup>r</sup>                                                                                                                                         | This work     |
| <i>cheY</i> -pME6032                                                                          | <i>cheY</i> overexpression vector in pME6032, <i>cheY</i> expression is controlled by P <sub>lac</sub> promoter; Tet <sup>r</sup>                                                                                                                                       | This work     |
| <i>yljH</i> -pME6032                                                                          | <i>yljH</i> overexpression vector in pME6032, <i>yljH</i> expression is controlled by P <sub>lac</sub> promoter; Tet <sup>r</sup>                                                                                                                                       | This work     |
| pCdrA- <i>gfp</i>                                                                             | pUCP22-NotI based cyclic di-GMP level reporter, Gm <sup>r</sup>                                                                                                                                                                                                         | Ref 45        |
| <i>cheY-eyfp</i> -pJN105                                                                      | <i>eyfp</i> fusions <i>cheY</i> cloned into pJN105, <i>cheY-eyfp</i> expression is controlled by P <sub>BAD</sub> promoter; Gm <sup>r</sup>                                                                                                                             | This work     |
| <i>cheA-ecfp</i> -pJN105                                                                      | <i>ecfp</i> fusions <i>cheA</i> cloned into pJN105, <i>cheA-ecfp</i> expression is controlled by P <sub>BAD</sub> promoter; Gm <sup>r</sup>                                                                                                                             | This work     |

**Table 1.**

**Strains and plasmids used in this study.**

bacterial 10-fold. Alexa Fluor 568 maleimide (Invitrogen-Molecular Probes) was added to a final concentration of 20  $\mu\text{g/ml}$ , and labeling was allowed to proceed for 30 min at room temperature with gyration at 80 rpm. Unused dye was then removed by washing the cells with MB three times, and the final pellet was resuspended in MB.

For fluorescence imaging, 50  $\mu\text{l}$  of cells were added to the chamber (constructed with two double-sticky tapes, a glass slide, and a poly-L-Lysine coated coverslip), incubated for 3 min, and rinsed with 100  $\mu\text{l}$  MB. The boundary of the chamber was then sealed with Apiezon vacuum grease. The chamber was placed on a Nikon Ti-E inverted fluorescence microscope with a 100 $\times$  oil-immersion objective and a sCMOS camera (Primer95B, Photometrics). The flagellar filament and chemotaxis complex of cells were observed separately with corresponding filter-set, using a 200-ms exposure time. To quantify the fluorescence intensity of the receptor cluster, we took the intensity maximum point as the center and extract a 3 $\times$ 3 pixel matrix around it. The mean value of the elements in this matrix represents the final fluorescence intensity.

## Western Blotting

The total protein of *P. aeruginosa* strain cells in each treatment group was extracted and its concentration was measured with Bicinchoninic acid (BCA Protein Quantification Kit, Yeasen Biotechnology Co., Ltd) when the cells grew to exponential phase with same reduced turbidity ( $\text{OD}_{600}$  between 0.9 and 1.0). The exact amount of protein was subjected to SDS-PAGE electrophoresis, then transferred to a nitrocellulose membrane, which was blocked with 50 mg/mL skim milk in TBST buffer (20 mM Tris, 150 mM NaCl [pH7.4], 0.1% Tween 20) for 1 h at room temperature. Since the N-terminal sequence of EYFP is almost identical to that of GFP, a rabbit anti-GFP antibody (1:2000, Yeasen Biotechnology Co., Ltd) and a Mouse anti- $\beta$ -actin antibody (1:2000, Invitrogen) were added and incubated at 4°C overnight. The nitrocellulose membrane was gently washed with TBST for 20 min, three times. Subsequently, HRP labeled goat anti-rabbit IgG (1:4000, Abcam) and rabbit anti-mouse IgG (1:4000, Abcam) were added and incubated at room temperature for 1 h. The nitrocellulose membrane was again gently washed with TBST for 20 min, three times. Then, the membrane was developed using the ECL chemiluminescence detection system (Beyotime P0018FS) and visualized with the ChemiDoc XRS+ system (Bio-Rad). Three independent experiments were conducted for reproducibility.

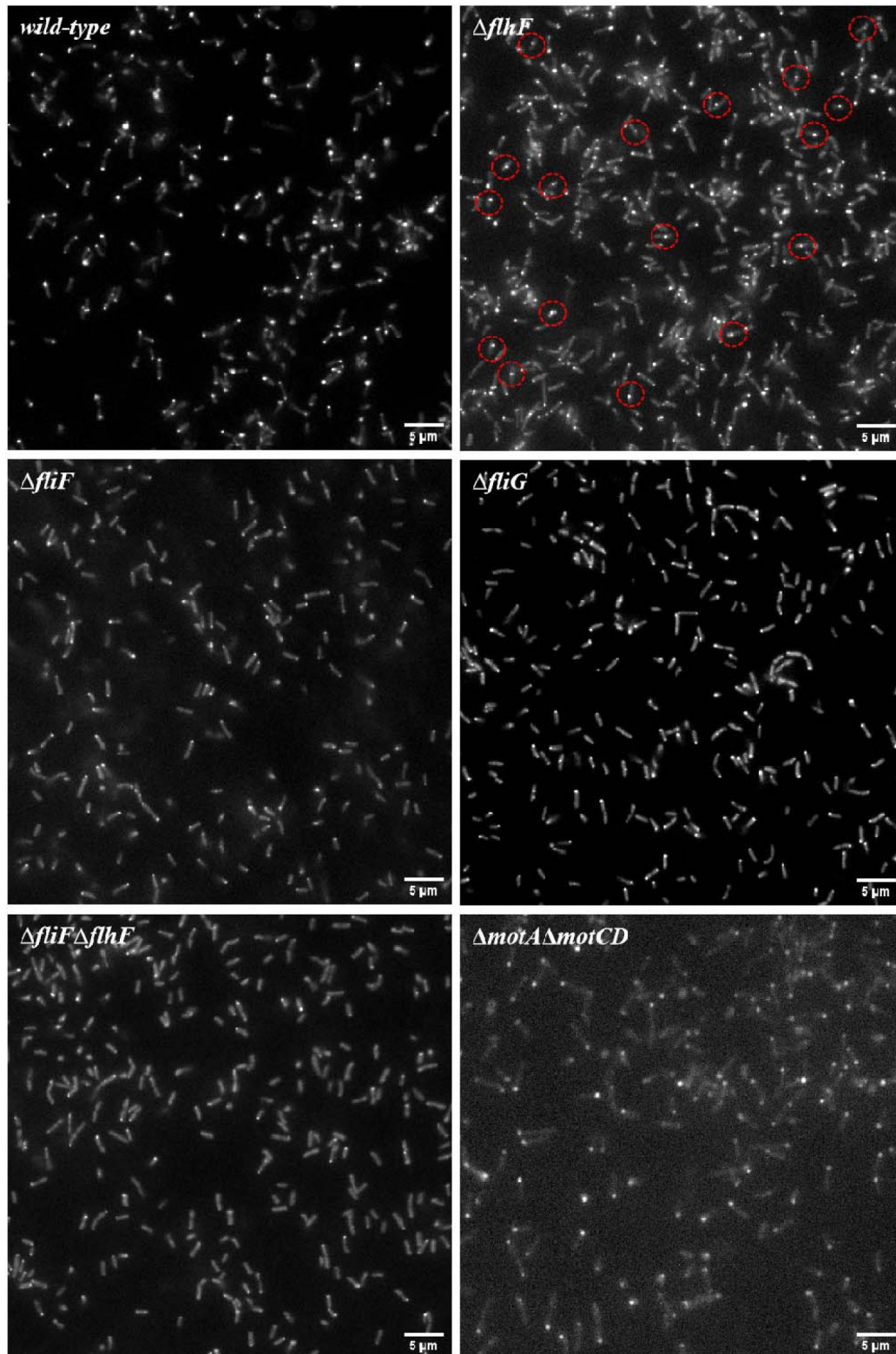
## Monitoring c-di-GMP signal at the single-cell level

For measurements of c-di-GMP signaling, the validated reporter plasmid pCdrA-*gfp*, which produces green fluorescent protein (GFP) in response to an increase in c-di-GMP, was introduced into the experimental system. A 1000 $\times$  diluted culture was inoculated into the apparatus described in the previous section and allowed to adhere to the surface for  $\sim 10$  min, with the slight difference that the coverslips were not coated with poly-L-Lysine to avoid interference from background fluorescence. Bacteria were imaged using a Nikon Ti-E inverted microscope, which was equipped with an EMCCD camera (DU897, iXon3, Andor Technology). We used a 100 $\times$  oil-immersion objective. Bacteria were illuminated with a 488-nm laser (Sapphire 488-200 mW, Coherent), using a standard GFP filter sets in the fluorescence imaging system with an exposure time of 500 ms. To ensure the consistency of culture and observation conditions, the data of wild-type and *cheY*-pME6032 carried strains were collected at the same time. More than 200 independent individuals of each strain were used for data analysis.

Single-cell c-di-GMP concentration was characterized by calculating fluorescence intensity. The average fluorescence intensity of a single cell was calculated by dividing the total fluorescence intensity of a single cell by the cell volume. Bacteria were simplified to a hollow cylindrical trunk with hollow hemispherical caps at both ends. The cell volume can be calculated by taking the length of the short axis of the cell body as the diameter of the sphere and the length of the long axis as the sum of the height of the cylinder and the diameter of the sphere. The mean

fluorescence intensity of the cell was obtained by subtracting the average background fluorescence intensity from the total fluorescence intensity of the cell, and then dividing by the cell volume.

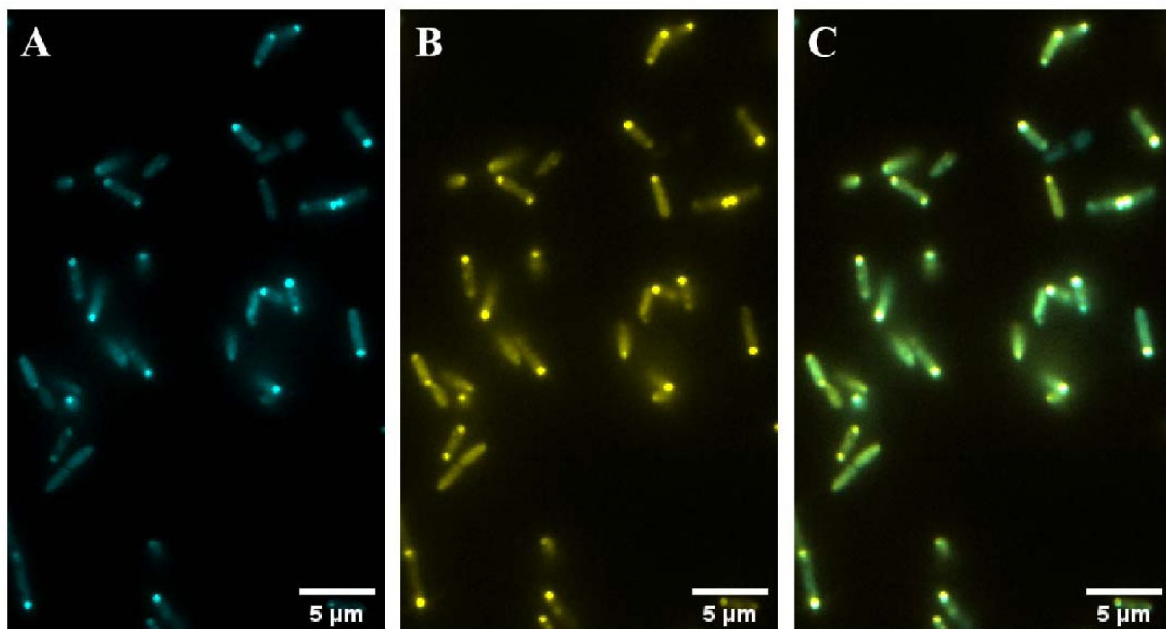
## Supplementary figures



**Fig. S1.**

**Distribution of chemotaxis complexes in multiple strains within representative large fields.**

Red dotted circles indicate cells where the chemotaxis complex is located at the mid-cell position.

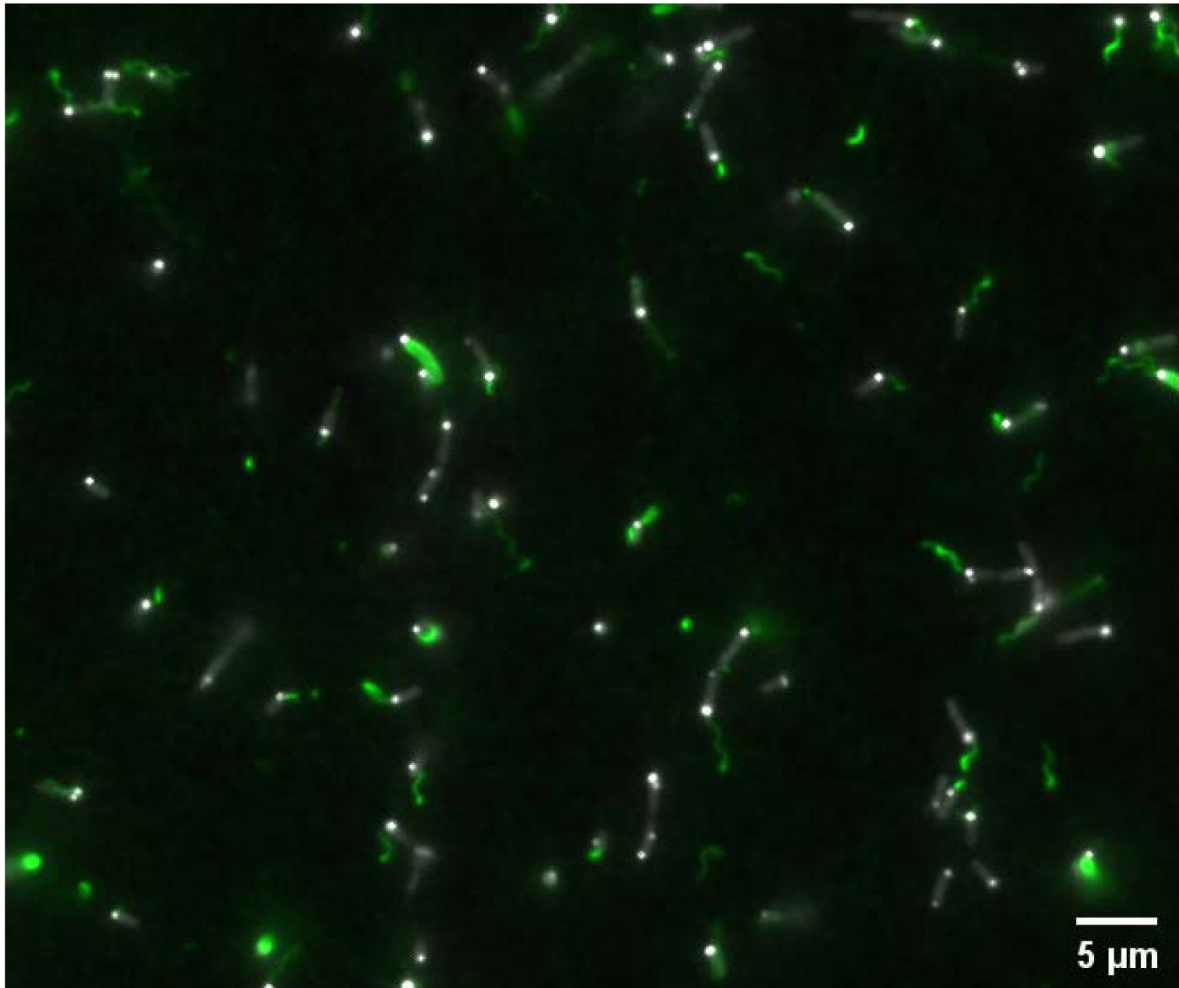


**Fig. S2.**

**Co-localization of CheY-EYFP and CheA-ECFP in *P. aeruginosa*.**

CheA-ECFP is shown in blue (left), CheY-EYFP in yellow (center), and the merged image is shown in the right panel.

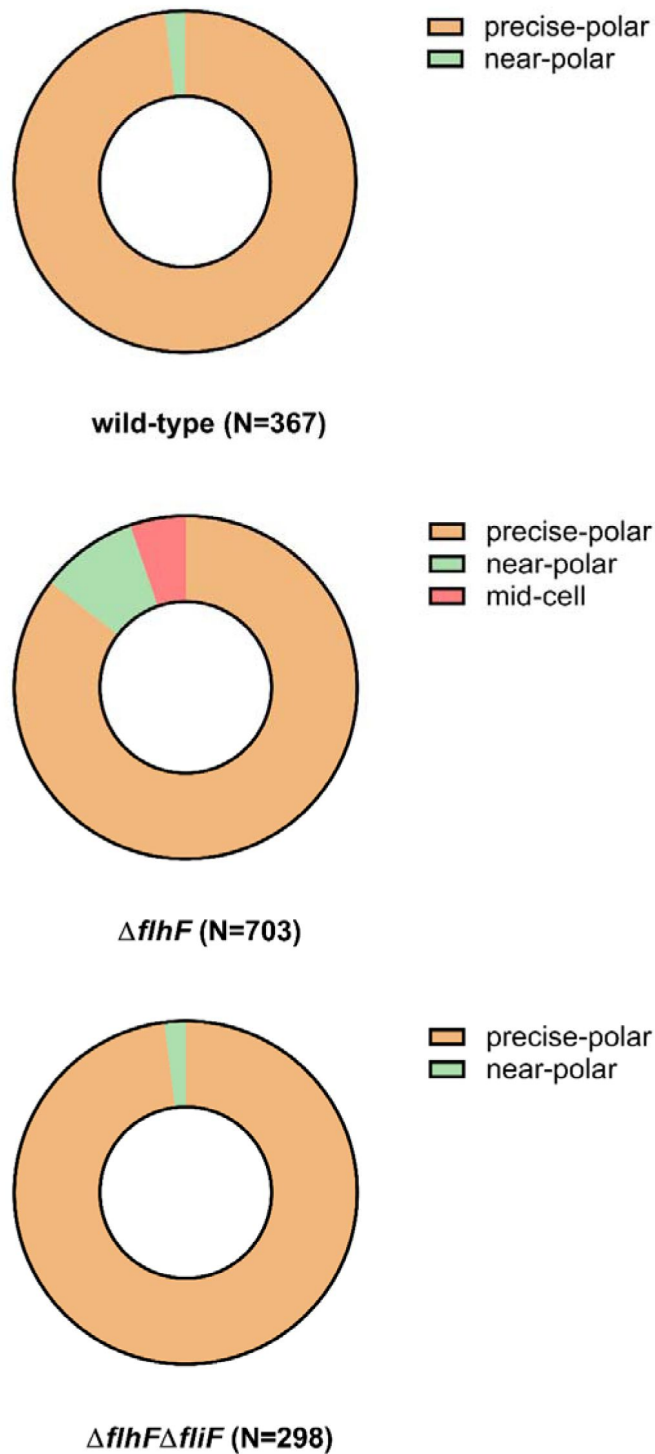




**Fig. S3.**

**Simultaneous observation of the chemotaxis complex and flagellar filaments in the wild-type strain, shown in a representative large field.**

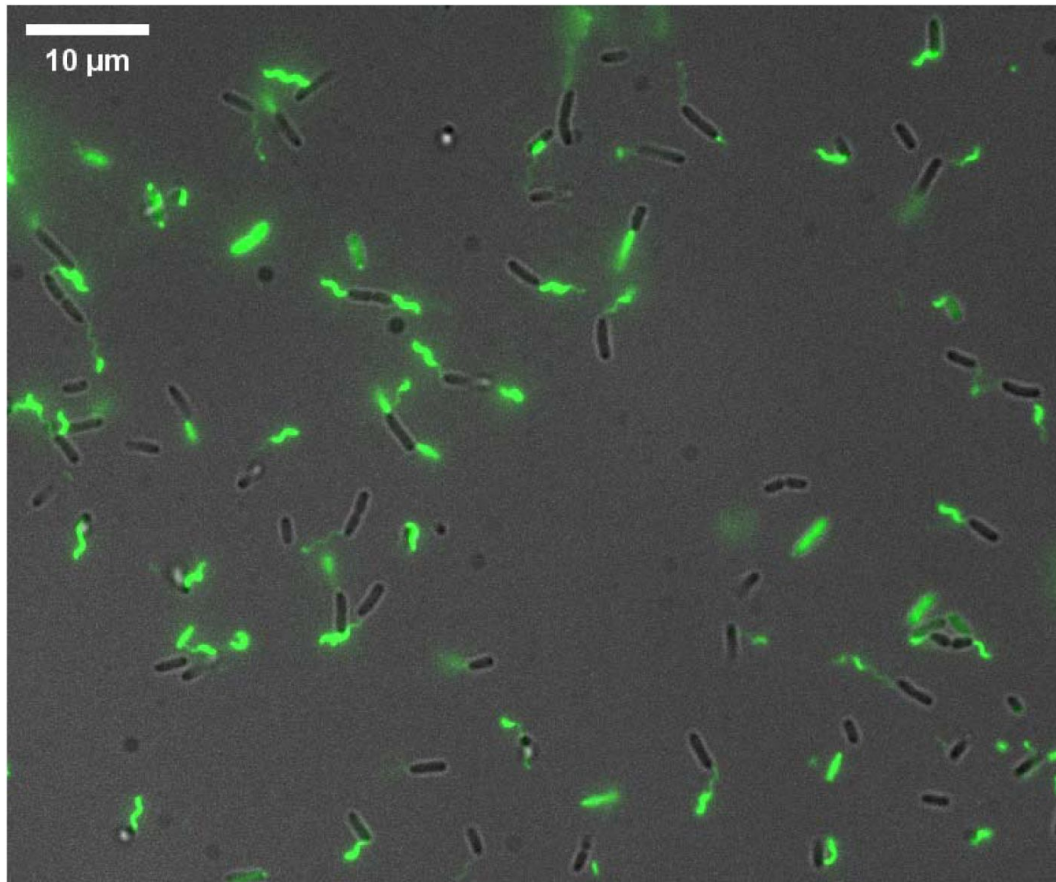




**Fig. S4.**

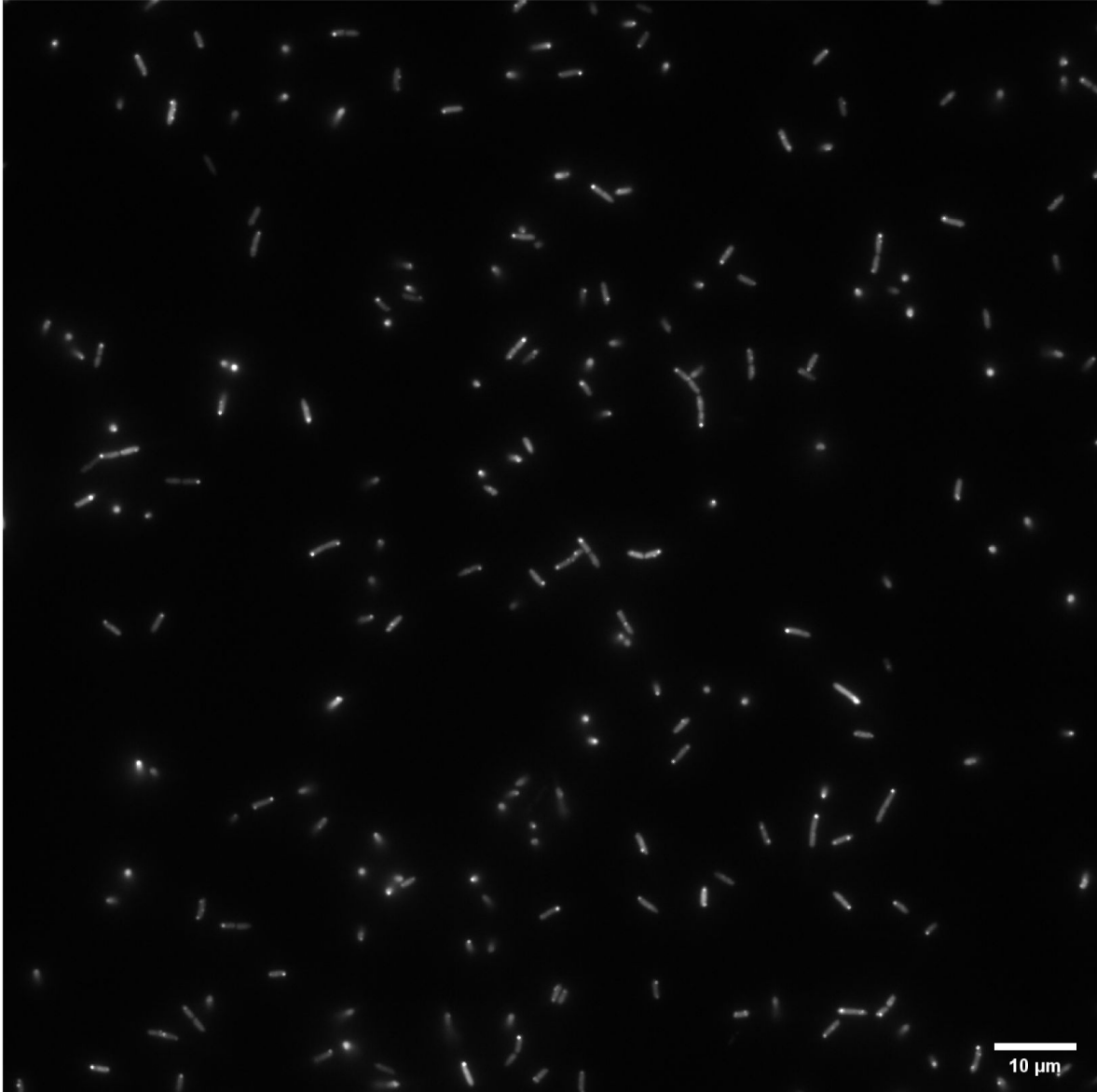
**Distribution statistics of the chemotaxis complex in the wild-type strain and the *flhF* mutant.**

The distribution patterns are categorized into three types: precise-polar, near-polar, and mid-cell localization. In the wild-type strain, 98.1% of cells exhibit precise-polar localization of the chemotaxis complex, with the remainder showing near-polar distribution. In the *flhF* mutant, the proportion of cells with precise-polar localization decreases to 85.5%, and mid-cell localization appears in approximately 5% of cells. The *flhF*-*fliF* double mutant displays a distribution pattern similar to that of the wild-type strain.



**Fig. S5.**

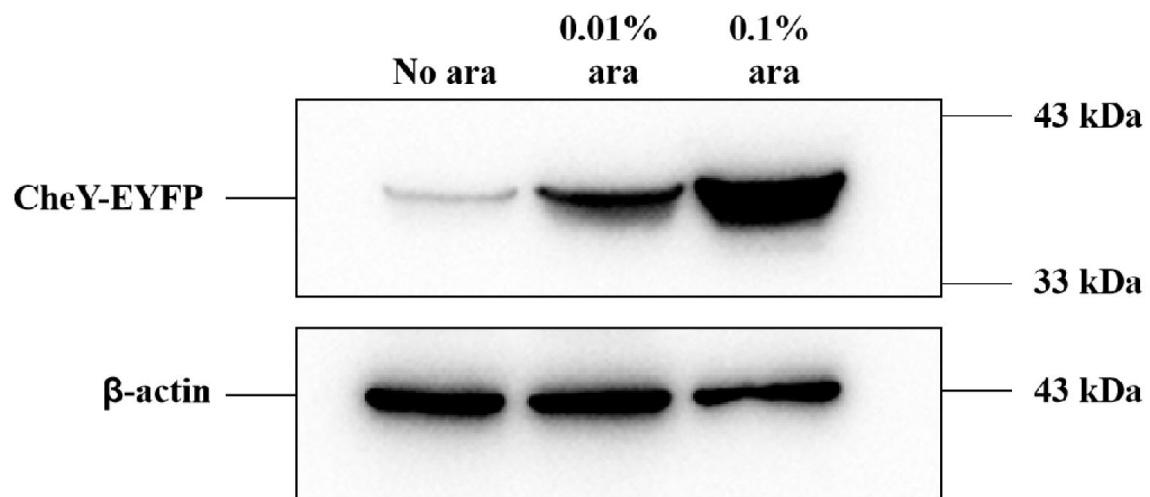
**Knockout of *cheA* did not affect lagellar assembly efficiency of *P. aeruginosa*.**



**Fig. S6.**

**Localization of CheY-EYFP in the  $\Delta flgI$  strain of *P. aeruginosa*.**

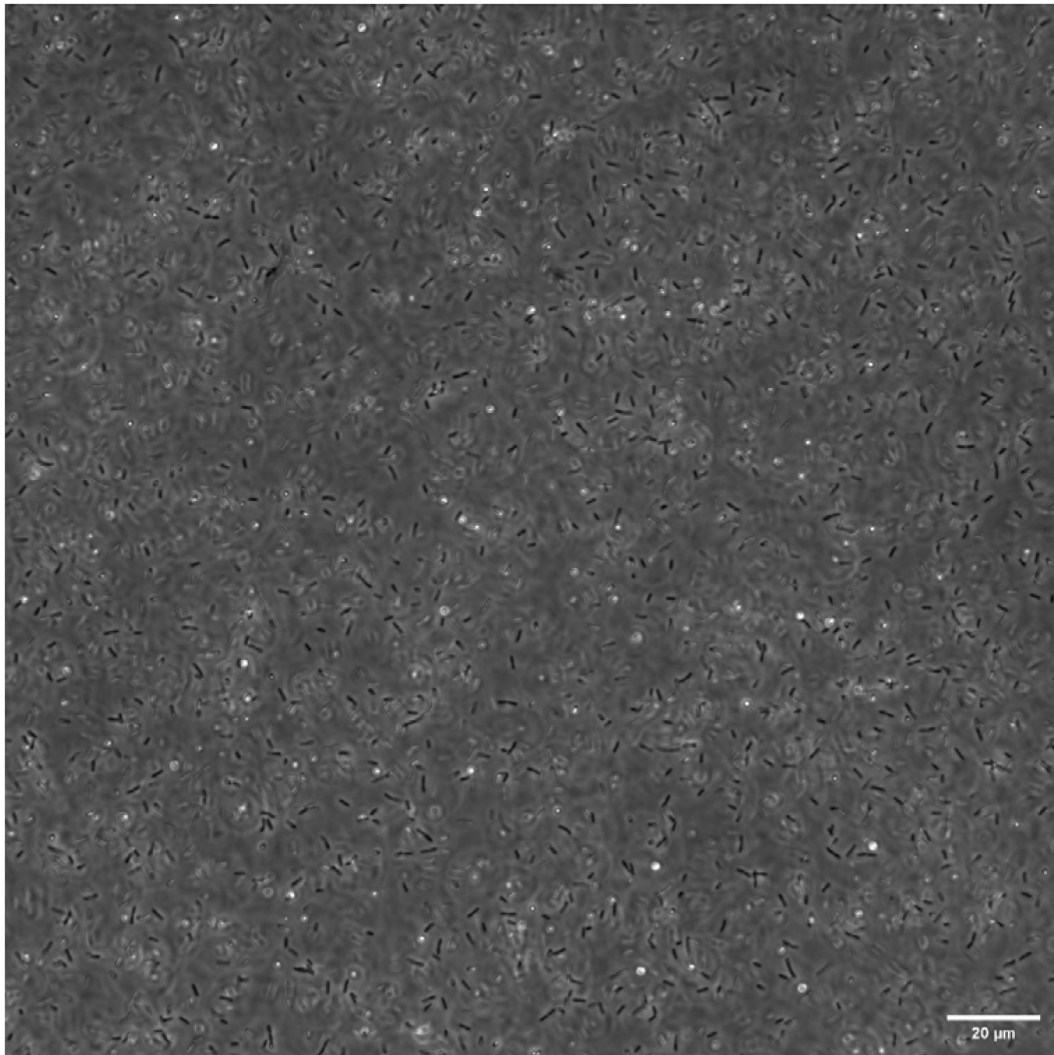
The phenotype is similar to that of the wild-type strain.



**Fig. S7.**

**Intracellular CheY levels induced by different arabinose concentrations, measured by western blot analysis, showing clear differences across the concentration gradient.**

β-actin was used as the housekeeping protein.



**Fig. S8.**

**Co-overexpressing CheY and the phosphodiesterase (PDE) YhjH from *E. coli* mitigates cell aggregation caused by CheY overexpression in *P. aeruginosa*.**



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## Additional information

### Author Contributions

J.Y. and R.Z. designed the work; Z.W., M.T. and S.F. performed the measurements with help from M.C.; Z.W. and S.F. analyzed the data; Z.W., J.Y. and R.Z. wrote the paper. Z.W., M.T. and S.F. contributed equally to this work.

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**Reviewer #1 (Public review):**

Summary:

The study by Wu et al presents interesting data on bacterial cell organization, a field that is progressing now, mainly due to the advances in microscopy. Based mainly on fluorescence microscopy images, the authors aim to demonstrate that the two structures that account for bacterial motility, the chemotaxis complex and the flagella, colocalize to the same pole in *Pseudomonas aeruginosa* cells and to expose the regulation underlying their spatial organization and functioning.

Comments on revisions:

The authors have addressed all major and minor points that I raised in a satisfying way during the revision process. The work can now be regarded as complete, the assumptions were clarified, the results are convincing, the conclusions are justified, and the novelty has been made clear.



This manuscript will be of interest to cell biologists, mainly those studying bacteria, but not only

<https://doi.org/10.7554/eLife.97514.2.sa3>

**Reviewer #2 (Public review):**

Summary:

Here, the authors studied the molecular mechanisms by which the chemoreceptor cluster and flagella motor of *Pseudomonas aeruginosa* (PA) are spatially organized in the cell. They argue that FlhF is involved in localizing the receptors-motor to the cell pole, and even without FlhF, the two are colocalized. Finally, the authors argue that the functional reason for this colocalization is to insulate chemotactic signaling from other signaling pathways, such as cyclic-di-GMP signaling.

Strength:

The experiments and data are high quality. It is clear that the motor and receptors co-localize, and that elevated CheY levels lead to elevated c-di-GMP.

Weakness:

The explanation for the functional importance of receptor-motor colocalization is plausible but is still not conclusively demonstrated. Colocalization might reduce CheY levels throughout the cell in order to reduce cross-talk with c-di-GMP. This would mean that if physiologically-relevant levels of CheYp near the pole were present throughout the cell, c-di-GMP levels would be elevated to a point that is problematic for the cell. Clearly demonstrating this seems challenging.

<https://doi.org/10.7554/eLife.97514.2.sa2>

**Reviewer #3 (Public review):**

Summary:

The authors investigated the assembly and polar localization of the chemosensory cluster in *P. aeruginosa*. They discovered that a certain protein (FlhF) is required for the polar localization of the chemosensory cluster while a fully-assembled motor is necessary for the assembly of the cluster. They found that flagella and chemosensory clusters always co-localize in the cell; either at the cell pole in wild type cells or randomly-located in the cell in FlhF mutant cells. They hypothesize that this co-localization is required to keep the level of another protein (CheY-P), which controls motor switching, at low levels as the presence of high-levels of this protein (if the flagella and chemosensory clusters were not co-localized) is associated with high-levels of c-di-GMP and cell aggregations.

Strengths:

The manuscript is clearly written and straightforward. The authors applied multiple techniques to study the bacterial motility system including fluorescence light microscopy and gene editing. In general, the work enhances our understanding of the subtlety of interaction between the chemosensory cluster and the flagellar motor to regulate cell motility.

Weaknesses:

The major weakness for me in this paper is that the authors never discussed how the flagellar genes expression is controlled in *P. aeruginosa*. For example, in *E. coli* there is a transcriptional hierarchy for the flagellar genes (early, middle, and late genes, see Chilcott and Hughes, 2000). Similarly, *Campylobacter* and *Helicobacter* have a different regulatory cascade for their flagellar genes (See Lertsethtakarn, Ottemann, and Hendrixson, 2011). How does the expression of flagellar genes in *P. aeruginosa* compare to other species? how many classes are there for these genes? is there a hierarchy in their expression and how does this affect the results of the *FliF* and *FliG* mutants? In other words, if *FliF* and *FliG* are in class I (as in *E. coli*) then their absence might affect the expression of other later flagellar genes in subsequent classes (i.e., chemosensory genes). Also, in both *FliF* and *FliG* mutants no assembly intermediates of the flagellar motor are present in the cell as *FliG* is required for the assembly of *FliF* (see Hiroyuki Terashima et al. 2020, Kaplan et al. 2019, Kaplan et al. 2022). It could be argued that when the motor is not assembled then this will affect the expression of the other genes (e.g., those of the chemosensory cluster) which might play a role in the decreased level of chemosensory clusters the authors find in these mutants.

Comments on revisions:

I believe the authors have performed additional experiments that improved their manuscript and they have answered many of my comments and those of the other reviewers. I am supportive of publishing this manuscript, but I still find the following points that are not clear to me (probably I am misunderstanding some points; the authors can clarify).

(1) In response to reviewer 1, the authors say that they "analyzed and categorized the distribution of the chemotaxis complex in both wild-type and *flhF* mutant strains into three patterns: precise-polar, near-polar, and mid-cell localization." I can see what they mean by polar and mid-cell, but near-polar sounds a bit elusive? Can they provide examples of this stage and mention how accurately they can identify it? Also, do the pie charts they show in Figure S4 really show "significant alterations"? There is a difference between 98% and 85% as they mention in their response to reviewer 1, but I am not sure that this is significant? Probably they can explain/change the language in the text? Also, the number of cells they counted for *FlhF* mutant is more than the double of other strains (WT and *FlhF* *FliF* mutant)?

(2) One thing that also confused me is the following: One point that the authors stress is that *FlhF* localizes both the flagellum and the chemoreceptors to the pole. However, if I look at Figure 2B, the flagellum and the chemoreceptors still co-localize together (although not at the pole). If *FlhF* was responsible for co-localizing both of them to the pole, then wouldn't one expect them to be randomly localized in this mutant and by that I mean that they do not co-localize but that each of them (the flagellum and the chemoreceptors) are located in a different random location of the cell (not co-localized). The fact that they are still co-localized together in this mutant could also be interpreted by, for example, that *FlhF* localizes the flagellum to the pole and another mechanism localizes the chemoreceptors to the flagellum, hence, they still co-localize in this mutant because the chemoreceptors follow the flagellum by another mechanism to wherever it goes?

(3) In the response to reviewers, the authors mention "suggesting that the assembly of the receptor complex is likely influenced mainly by the C-ring and MS-ring structures rather than by the P ring". However, in the article, they still write "The complete assembly of the motor serves as a partial prerequisite for the assembly of the chemotaxis complex, and its assembly site is also regulated by the polar anchor protein *FlhF*" despite their *FlgI* results which is not in accordance with this statement? Also, As I mentioned in my previous report, in *FliG* and *FliF* mutant the motor does not assemble (see Hiroyuki Terashima et al. 2020., and Kaplan et al., 2022).

(4) The authors have said in their response to my point "and currently, there is no evidence that FliA activity is influenced by proteins like FliG". I just want to clarify what I meant in my previous report: In *E. coli*, FliA binds to FlgM, and when the hook is assembled FlgM is secreted outside the cell allowing FliA to trigger the transcription of class III genes, which include the chemosensory genes (see Figure 5 in Beeby et al, 2020 in FEMS Microbiology, and Chilcott and Hughes, 2000). This implies that if the hook is not built, then late genes (including the chemoreceptors) should not be present. However, in Kaplan et al., 2019, the authors imaged a FliF mutant in *Shewanella oneidensis* (Figure S3) and still saw that chemoreceptors are present (I believe the authors must highlight this). This suggests that species such as *Shewanella* and *Pseudomonas* have a different assembly process than that *E. coli*, and although the authors say that in the text, I believe they still can refine this part more in the spirit of what I wrote here.

I do not like to ask for additional experiments in the second round of review, so for me if the authors modify the text to tackle these points and allow for probable alternative explanations/ highlight gaps/ modify language used for some claims, then that is fine with me.

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#### Author response:

The following is the authors' response to the original reviews.

##### **Public Reviews:**

##### **Reviewer #1 (Public Review):**

###### *Summary:*

*The study by Wu et al presents interesting data on bacterial cell organization, a field that is progressing now, mainly due to the advances in microscopy. Based mainly on fluorescence microscopy images, the authors aim to demonstrate that the two structures that account for bacterial motility, the chemotaxis complex and the flagella, colocalize to the same pole in *Pseudomonas aeruginosa* cells and to expose the regulation underlying their spatial organization and functioning.*

###### *Strengths:*

*The subject is of importance.*

###### *Weaknesses:*

*The conclusions are too strong for the presented data. The lack of statistical analysis makes this paper incomplete. The novelty of the findings is not clear.*

We have strengthened the data analysis by including appropriate statistical tests to support our conclusions more convincingly. Additionally, we have refined the description of the research background to better emphasize the novelty and significance of our findings. Please see the detailed responses below for further information.

###### *Major issues:*

*(1) The novelty is in question since in the Abstract the authors highlight their main finding, which is that both the chemotaxis complex and the flagella localize to the same pole, as surprising. However, in the Introduction they state that "pathway-related receptors that mediate chemotaxis, as well as the flagellum are localized at the same cell*

*pole17,18". I am not a pseudomonas researcher and from my short glance at these references, I could not tell whether they report colocalization of the two structures to the same pole. However, I trust the authors that they know the literature on the localization of the chemotaxis complex and flagella in their organism. See also major issue number 5 on the novelty regarding the involvement of c-di-GMP.*

We thank the reviewer for this valuable comment and appreciate the opportunity to clarify our statements.

Kazunobu et al. (ref. 18) used scanning electron microscopy to preliminarily characterize the flagellation pattern of *Pseudomonas aeruginosa* during cell division, showing that existing flagella are located at the old pole. Zehra et al. (ref. 17), through fluorescence microscopy, observed that CheA and CheY proteins in dividing cells are typically also present at the old pole. Based on these observations, we inferred in the Introduction that the chemotaxis complex and flagellum may localize to the same cell pole.

However, this inference is indirect and lacks direct live-cell evidence of colocalization, leaving its validity to be confirmed. This uncertainty was indeed the starting point and motivation for our study.

In our work, we simultaneously visualized flagellar filaments and core chemoreceptor proteins at the single-cell level in *P. aeruginosa*. We characterized the assembly and spatial coordination of the chemotaxis network and flagellar motor throughout the cell cycle, providing direct evidence of their colocalization and coordinated assembly. This represents a significant advance beyond prior indirect observations and supports the novelty of our study.

Accordingly, we have revised the relevant statements in lines 71-75 of the manuscript to better reflect the current state of the literature and emphasize the novelty of our direct observations.

*(2) Statistics for the microscopy images, on which most conclusions in this manuscript are based, are completely missing. Given that most micrographs present one or very few cells, together with the fact that almost all conclusions depend on whether certain macromolecules are at one or two poles and whether different complexes are in the same pole, proper statistics, based on hundreds of cells in several fields, are absolutely required. Without this information, the results are anecdotal and do not support the conclusions. Due to the importance of statistics for this manuscript, strict statistical tests should be used and reported. Moreover, representative large fields with many cells should be added as supportive information.*

We thank the reviewer for this important comment, which significantly improves the rigor and persuasiveness of our manuscript.

For the colocalization analyses presented in Fig. 1D and Fig. 2B, we quantified 145 and 101 cells with fluorescently labeled flagella, respectively, and observed consistent colocalization of the chemoreceptor complexes and flagella in all examined cells (now added in the figure legends). Regarding the distribution patterns of chemoreceptors shown in Fig. 3A, we have now included comprehensive statistical analyses for both wild-type and mutant strains. For each strain, more than 300 cells were analyzed across at least three independent microscopic fields, providing robust statistical power (detailed data are presented in Fig. 3C).

To further strengthen the evidence, statistical tests were applied to confirm the significance and reproducibility of our findings (Fig. 3C). In addition, representative large-field fluorescence images containing numerous cells have been added to the supplementary materials (Fig. S1 and Fig. S3).

*The problem is more pronounced when the authors make strong statements, as in lines 157-158: "The results revealed that the chemoreceptor arrays no longer grow robustly at the cell pole (Figure 2A)". Looking at the seven cells shown in Figure 2A, five of them show polar localization of the chemoreceptors. The question is then: what is the percentage of cells that show precise polar, near-polar, or mid cell localization (the three patterns shown here) in the mutant and in the wild type? Since I know that these three patterns can also be observed in WT cells, what counts is the difference, and whether it is statistically significant.*

We thank the reviewer for raising this important point. Following the reviewer's suggestion, we have now analyzed and categorized the distribution of the chemotaxis complex in both wild-type and  $\Delta fliH$  mutant strains into three patterns: precise-polar, near-polar, and mid-cell localization. For each strain, more than 200 cells across three independent fields of view were quantified.

Our statistical analysis shows that in the wild-type strain, approximately 98% of cells exhibit precise polar localization of the chemotaxis complex. In contrast, the  $\Delta fliH$  mutant displays a clear shift in distribution, with about 5% of cells showing mid-cell localization and 9.5% showing near-polar localization. These differences demonstrate a significant alteration in the spatial pattern upon  $fliH$  deletion.

We have revised the relevant text in lines 166-170 accordingly and included the detailed statistical data in the newly added Fig. S4.

*Even for the graphs shown in Figures 3C and 3D, where the proportion of cells with obvious chemoreceptor arrays and absolute fluorescence brightness of the chemosensory array are shown, respectively, the questions that arise are: for how many individual cells these values hold and what is the significance of the difference between each two strains?*

The number of cells analyzed for each strain is indicated in the original manuscript: 372 wild-type cells (line 123), 221  $\Delta fliH$  cells (line 172), 234  $\Delta fliG$  cells (line 197), 323  $\Delta fliF$  cells (line 200), 672  $\Delta fliH\Delta fliF$  cells (line 202), and 242  $\Delta motA\Delta motCD$  cells (line 207). For each strain, data were collected from three independent fields of view. We have now also provided the number of cells in Fig. 3 legend.

We have now performed statistical comparisons using t-tests between strains. Notably, the measured values in Fig. 3C exhibit a clear, monotonic decrease with successive gene knockouts, supporting the robustness of the observed trend.

Regarding the absolute fluorescence intensity shown in the original Fig. 3D, the mutants did not display consistent directional changes compared to the wild type. Reliable comparison of absolute fluorescence intensity requires consistent fluorescent protein maturation levels across strains. Given the likely variability in maturation levels between strains, we concluded that this data may not accurately reflect true differences in protein concentrations. Therefore, we have removed the fluorescence intensity graph from the revised manuscript to avoid potential misinterpretation.

*(3) The authors conclude that "Motor structural integrity is a prerequisite for chemoreceptor self-assembly" based on the reduction in cells with chemoreceptor clusters in mutants deleted for flagellar genes, despite the proper polar localization of the chemotaxis protein CheY. They show that the level of CheY in the WT and the mutant strains is similar, based on Western blot, which in my opinion is over-exposed. "To ascertain whether it is motor integrity rather than functionality that influences the*

*efficiency of chemosensory array assembly", they constructed a mutant deleted for the flagella stator and found that the motor is stalled while CheY behaves like in WT cells. The authors further "quantified the proportion of cells with receptor clusters and the absolute fluorescence intensity of individual clusters (Figures 3C-D)". While Figure 3DC suggests that, indeed, the flagella mutants show fewer cells with a chemotaxis complex, Figure 3D suggests that the differences in fluorescence intensity are not statistically significant. Since it is obvious that the regulation of both structures' production and localization is codependent, I think that it takes more than a Western blot to make such a decision.*

We thank the reviewer for the suggestions. To further clarify that the assembly of flagellar motors and chemoreceptor clusters occurs in an orderly manner rather than being merely codependent, we performed additional experiments. Specifically, we constructed a  $\Delta cheA$  mutant strain, in which chemoreceptor clusters fail to assemble. Using in vivo fluorescent labeling of flagellar filaments, we observed that the proportion of cells with flagellar filaments in the  $\Delta cheA$  strain was comparable to that of the wild type (Fig. S5).

In contrast, mutants lacking complete motor structures, such as  $\Delta fliF$  and  $\Delta fliG$ , showed a significant reduction in the proportion of cells with obvious receptor clusters (Fig. 3C). Based on these results, we conclude that the structural integrity of the flagellar motor is, to a certain extent, a prerequisite for the self-assembly of chemoreceptor clusters.

Accordingly, we have revised the relevant statement in lines 213-217 of the manuscript to reflect this clarification.

*(4) I wonder why the authors chose to label CheY, which is the only component of the chemotaxis complex that shuttles back and forth to the base of the flagella. In any case, I think that they should strengthen their results by repeating some key experiments with labeled CheW or CheA.*

We thank the reviewer for this valuable suggestion. In our study, we initially focused on the positional relationship between chemoreceptor clusters and flagella, then investigated factors influencing cluster distribution and assembly efficiency. The physiological significance of motor and cluster co-localization was ultimately proposed with CheY as the starting point.

Previous work by Harwood's group demonstrated that both CheY-YFP and CheA-GFP localize to the old poles of dividing *Pseudomonas aeruginosa* cells. Since our physiological hypothesis centers on CheY, we chose to label CheY-EYFP in our experiments.

To further strengthen our conclusions, we constructed a plasmid expressing CheA-CFP and introduced it into the *cheY-eyfp* strain via electroporation. Fluorescence imaging revealed a high degree of spatial overlap between CheA-CFP and CheY-EYFP (Fig. S2), confirming that CheY-EYFP accurately marks the location of the chemoreceptor complex.

We have revised the manuscript accordingly (lines 119-123) and added these data as Fig. S2.

*(5) The last section of the results is very problematic, regarding the rationale, the conclusions, and the novelty. As far as the rationale is concerned, I do not understand why the authors assume that "a spatial separation between the chemoreceptors and flagellar motors should not significantly impact the temporal comparison in bacterial chemotaxis". Is there any proof for that?*

We apologize for the lack of clarity in our original explanation. The rationale behind the statement was initially supported by comparing the timescales of CheY-P diffusion and temporal comparison in chemotaxis. Specifically, the diffusion time for CheY-P to traverse the



entire length of a bacterial cell is approximately 100 ms (refs 39&40), whereas the timescale for bacterial chemotaxis temporal comparison is on the order of seconds (ref 41).

To clarify and strengthen this argument, we have expanded the discussion as follows:

The diffusion coefficient of CheY in bacterial cells is about 10  $\mu\text{m}^2/\text{s}$ , which corresponds to an estimated end-to-end diffusion time on the order of 100 ms (refs 40&41). If the chemotaxis complexes were randomly distributed rather than localized, diffusion times would be even shorter. In contrast, the timescale for the chemotaxis temporal comparison is on the order of seconds (ref. 42). Additionally, a study by Fukuoka and colleagues reported that intracellular chemotaxis signal transduction requires approximately 240 ms beyond CheY or CheY-P diffusion time (ref. 41). Moreover, the intervals of counterclockwise (CCW) and clockwise (CW) rotation of the *P. aeruginosa* flagellar motor under normal conditions are 1-2 seconds, as determined by tethered cell or bead assays (refs. 30&43).

Taken together, these indicate that for *P. aeruginosa*, which moves via a run-reverse mode, the potential 100 ms reduction in response time due to co-localization of the chemotaxis complex and motor has a limited effect on overall chemotaxis timing.

We have revised the corresponding text accordingly (lines 238-245) to better explain this rationale.

*More surprising for me was to read that "The signal transduction pathways in E. coli are relatively simple, and the chemotaxis response regulator CheY-P affects only the regulation of motor switching". There are degrees of complexity among signal transduction pathways in E. coli, but the chemotaxis seems to be ranked at the top. CheY is part of the adaptation. Perfect adaptation, as many other issues related to the chemotaxis pathway, which include the wide dynamic range, the robustness, the sensitivity, and the signal amplification (gain), are still largely unexplained. Hence, such assumptions are not justified.*

We apologize for the confusion and imprecision in our original statements. Our intention was to convey that the chemotaxis pathway in *E. coli* is relatively simple compared to the more complex chemosensory systems in *P. aeruginosa*. We did not mean to generalize this simplicity to all signal transduction pathways in *E. coli*.

We acknowledge that *E. coli* chemotaxis is a highly sophisticated system, involving processes such as perfect adaptation, wide dynamic range, robustness, sensitivity, and signal amplification, many aspects of which remain incompletely understood. CheY indeed plays a crucial role in adaptation and motor switching regulation.

Accordingly, we have revised the original text (lines 249-255) to avoid any misunderstanding.

*More perplexing is the novelty of the authors' documentation of the effect of the chemotaxis proteins on the c-di-GMP level. In 2013, Kulasekara et al. published a paper in eLife entitled "c-di-GMP heterogeneity is generated by the chemotaxis machinery to regulate flagellar motility". In the same year, Kulasekara published a paper entitled "Insight into a Mechanism Generating Cyclic di-GMP Heterogeneity in Pseudomonas aeruginosa". The authors did not cite these works and I wonder why.*

We apologize for having been unaware of these important references and thank the reviewer for bringing them to our attention. We have now cited the eLife paper and the PhD thesis titled "Insight into a Mechanism Generating Cyclic di-GMP Heterogeneity in *Pseudomonas aeruginosa*" by Kulasekara et al.

Regarding novelty, there are key differences between our findings and those reported by Kulasekara et al. While they proposed that CheA influences c-di-GMP heterogeneity through interaction with a specific phosphodiesterase (PDE), our results demonstrate that overexpression of CheY leads to an increase in intracellular c-di-GMP levels.

We have revised the original text accordingly (lines 358-362) to clarify these distinctions.

*(6) Throughout the manuscript, the authors refer to foci of fluorescent CheY as "chemoreceptor arrays". If anything, these foci signify the chemotaxis complex, not the membrane-traversing chemoreceptors.*

We thank the reviewer for this clarification. We have revised the manuscript accordingly to refer to the fluorescent CheY foci as representing the chemotaxis complex rather than the chemoreceptor arrays.

#### *Conclusions:*

*The manuscript addresses an interesting subject and contains interesting, but incomplete, data.*

#### **Reviewer #2 (Public Review):**

##### *Summary:*

*Here, the authors studied the molecular mechanisms by which the chemoreceptor cluster and flagella motor of *Pseudomonas aeruginosa* (PA) are spatially organized in the cell. They argue that FlhF is involved in localizing the receptors-motor to the cell pole, and even without FlhF, the two are colocalized. FlhF is known to cause the motor to localize to the pole in a different bacterial species, *Vibrio cholera*, but it is not involved in receptor localization in that bacterium. Finally, the authors argue that the functional reason for this colocalization is to insulate chemotactic signaling from other signaling pathways, such as cyclic-di-GMP signaling.*

##### *Strengths:*

*The experiments and data look to be high-quality.*

##### *Weaknesses:*

*However, the interpretations and conclusions drawn from the experimental observations are not fully justified in my opinion.*

*I see two main issues with the evidence provided for the authors' claims.*

##### *(1) Assumptions about receptor localization:*

*The authors rely on YFP-tagged CheY to identify the location of the receptor cluster, but CheY is a diffusible cytoplasmic protein. In *E. coli*, CheY has been shown to localize at the receptor cluster, but the evidence for this in PA is less strong. The authors refer to a paper by Guvener et al 2006, which showed that CheY localizes to a cell pole, and CheA (a receptor cluster protein) also localizes to a pole, but my understanding is that colocalization of CheY and CheA was not shown. My concern is that CheY could instead localize to the motor in PA, say by binding FlhM. This "null model" would explain the authors' observations, without colocalization of the receptors and motor. Verifying that CheY and CheA are colocalized in PA would be a very helpful experiment to address this weakness.*

We thank the reviewer for this valuable suggestion. We agree that verifying the colocalization of CheY and CheA would strengthen our conclusions. To address this, we constructed a plasmid expressing CheA-CFP and introduced it into the CheY-EYFP strain by electroporation. Fluorescence imaging revealed a high degree of spatial overlap between CheA-CFP and CheY-EYFP signals, indicating that CheY-EYFP indeed marks the location of the chemoreceptor complex rather than the flagellar motor.

We have revised the manuscript accordingly (lines 118-123) and included these results in the new Fig. S2.

*(2) Argument for the functional importance of receptor-motor colocalization at the pole:*

*The authors argue that colocalization of the receptors and motors at the pole is important because it could keep phosphorylated CheY, CheY-p, restricted to a small region of the cell, preventing crosstalk with other signaling pathways. Their evidence for this is that overexpressing CheY leads to higher intracellular cdG levels and cell aggregation. Say that the receptors and motors are colocalized at the pole. In *E. coli*, CheY-p rapidly diffuses through the cell. What would prevent this from occurring in PA, even with colocalization?*

We appreciate the reviewer's insightful question. The colocalization of both the signaling source (the kinase) and sink (the phosphatase) at the chemoreceptor complex at the cell pole results in a rapid decay of CheY-P concentration within approximately 0.2  $\mu\text{m}$  from the cell pole, leading to a nearly uniform distribution elsewhere in the cell, as demonstrated by Vaknin and Berg (ref. 46). This spatial arrangement effectively confines high CheY-P levels to the pole region. When the motor is also localized at the cell pole, this reduces the need for elevated CheY-P concentrations throughout the cytoplasm, thereby minimizing potential crosstalk with other signaling pathways.

We have revised the manuscript accordingly (lines 280-286) to clarify this point.

*Elevating CheY concentration may increase the concentration of CheY-p in the cell, but might also stress the cells in other unexpected ways. It is not so clear from this experiment that elevated CheY-p throughout the cell is the reason that they aggregate, or that this outcome is avoided by colocalizing the receptors and motor at the same pole. If localization of the receptor array and motor at one pole were important for keeping CheY-p levels low at the opposite pole, then we should expect cells in which the receptors and motor are not at the pole to have higher CheY-p at the opposite pole. According to the authors' argument, it seems like this should cause elevated cdG levels and aggregation in the delta flhF mutants with wild-type levels of CheY. But it does not look like this happened. Instead of varying CheY expression, the authors could test their hypothesis that receptor-motor colocalization at the pole is important for preventing crosstalk by measuring cdG levels in the flhF mutant, in which the motor (and maybe the receptor cluster) are no longer localized in the cell pole.*

We thank the reviewer for raising the important point regarding potential cellular stress caused by elevated CheY concentrations, as well as for the suggestion to test the hypothesis using  $\Delta\text{flhF}$  mutants.

First, as noted above, CheY-P concentration rapidly decreases away from the receptor complex. While deletion of *flhF* alters the position of the receptor complex, thereby shifting the region of high CheY-P concentration, it does not increase CheY-P levels elsewhere in the cell. Importantly, in the  $\Delta\text{flhF}$  strain, the receptor complex and the motor still colocalize, so this mutant may not effectively test the role of receptor-motor colocalization in preventing crosstalk as suggested.

Regarding the possibility that elevated CheY levels stress the cells independently of CheY-P signaling, prior work in *E. coli* by Cluzel et al. (ref. 11) showed that overexpressing CheY several-fold did not cause phenotypic changes, indicating that simple CheY overexpression alone may not be generally stressful. Furthermore, our data indicate that the increase in c-di-GMP levels and subsequent cell aggregation upon CheY overexpression is not an all-or-none switch but occurs progressively as CheY concentration rises.

To further confirm that CheY overexpression promotes aggregation through increased c-di-GMP levels, we performed additional experiments co-overexpressing CheY and a phosphodiesterase (PDE) from *E. coli* to reduce intracellular c-di-GMP. These experiments showed that PDE expression mitigates cell aggregation caused by CheY overexpression (Fig. S8).

We have revised the manuscript accordingly (lines 290-294) and added these new results in Fig. S8.

### **Reviewer #3 (Public Review):**

#### *Summary:*

*The authors investigated the assembly and polar localization of the chemosensory cluster in P. aeruginosa. They discovered that a certain protein (FlhF) is required for the polar localization of the chemosensory cluster while a fully-assembled motor is necessary for the assembly of the cluster. They found that flagella and chemosensory clusters always co-localize in the cell; either at the cell pole in wild-type cells or randomly-located in the cell in FlhF mutant cells. They hypothesize that this co-localization is required to keep the level of another protein (CheY-P), which controls motor switching, at low levels as the presence of high levels of this protein (if the flagella and chemosensory clusters were not co-localized) is associated with high-levels of c-di-GMP and cell aggregations.*

#### *Strengths:*

*The manuscript is clearly written and straightforward. The authors applied multiple techniques to study the bacterial motility system including fluorescence light microscopy and gene editing. In general, the work enhances our understanding of the subtlety of interaction between the chemosensory cluster and the flagellar motor to regulate cell motility.*

#### *Weaknesses:*

*The major weakness in this paper is that the authors never discussed how the flagellar gene expression is controlled in P. aeruginosa. For example, in E. coli there is a transcriptional hierarchy for the flagellar genes (early, middle, and late genes, see Chilcott and Hughes, 2000). Similarly, Campylobacter and Helicobacter have a different regulatory cascade for their flagellar genes (See Lertsethtakarn, Ottemann, and Hendrixson, 2011). How does the expression of flagellar genes in P. aeruginosa compare to other species? How many classes are there for these genes? Is there a hierarchy in their expression and how does this affect the results of the FliF and FliG mutants? In other words, if FliF and FliG are in class I (as in E. coli) then their absence might affect the expression of other later flagellar genes in subsequent classes (i.e., chemosensory genes). Also, in both FliF and FliG mutants no assembly intermediates of the flagellar motor are present in the cell as FliG is required for the assembly of FliF (see Hiroyuki Terashima et al. 2020, Kaplan et al. 2019, Kaplan et al. 2022). It could be argued that when the motor is not assembled then this will affect the expression of the other genes (e.g., those of the chemosensory cluster) which might play a role in the decreased level of chemosensory clusters the authors find in these mutants.*

We thank the reviewer for the insightful comments. *P. aeruginosa* possesses a four-tiered transcriptional regulatory hierarchy controlling flagellar biogenesis. Within this system, *fliF* and *fliG* belong to class II genes and are regulated by the master regulator FleQ. In contrast, chemotaxis-related genes such as *cheA* and *cheW* are regulated by intracellular free FliA, and currently, there is no evidence that FliA activity is influenced by proteins like FliG.

To verify that the expression of core chemotaxis proteins was not affected by deletion of *fliG*, we performed Western blot analyses to compare CheY levels in wild-type,  $\Delta fliF$ , and  $\Delta fliG$  strains. We observed no significant differences, indicating that the reduced presence of receptor clusters in these mutants is not due to altered expression of chemotaxis proteins.

Accordingly, we have revised the manuscript (lines 341-348) and updated Fig. 3B to reflect these findings.

#### **Recommendations for the authors:**

##### **Reviewing Editor (Recommendations For The Authors):**

*The reviewers comment on several important aspects that should be addressed, namely: the lack of statistical analysis; the need for clarifications regarding assumptions made regarding receptor localization; the functional importance of receptor-motor colocalization; and the need for an elaborate discussion of flagellar gene expression. Also, two reviewers pointed out the need to prove the co-localization of CheY and CheA; This is important since CheY is dynamic, shuttling back and forth from the chemotaxis complex to the base of the flagella, whereas CheA (or cheW or, even better, the receptors) is considered less dynamic and an integral part of the chemotaxis complex.*

##### **Reviewer #1 (Recommendations For The Authors):**

*Minor points:*

*Line 43: "ubiquitous" - I would choose another word.*

We changed "ubiquitous" to "widespread".

*Line 49: "order" - change to organize.*

We changed "order" to "organize".

*Line 52: "To grow and colonize within the host, bacteria have evolved a mechanism for migrating...". Motility "towards more favorable environments" is an important survival strategy of bacteria in various ecological niches, not only within the host.*

We revised it to "grow and colonize in various ecological niches".

*Line 72: Define F6 in "F6 pathway-related receptors".*

The proteins encoded by chemotaxis-related genes collectively constitute the F6 pathway, which we have now explained in the manuscript text.

*Line 72-73: Do references 17 & 18 really report colocalization of the chemotaxis receptor and flagella to the same pole? If these or other reports document such colocalization, then the sentence in the Abstract "Surprisingly, we found that both are located at the same cell pole..." is not correct.*

Kazunobu et al. (ref. 18) used scanning electron microscopy to preliminarily characterize the flagellation pattern of *Pseudomonas aeruginosa* during cell division, showing that existing flagella are located at the old pole. Zehra et al. (ref. 17), through fluorescence microscopy, observed that CheA and CheY proteins in dividing cells are typically also present at the old pole. Based on these observations, we inferred in the Introduction that the chemotaxis complex and flagellum may localize to the same cell pole.

However, this inference is indirect and lacks direct live-cell evidence of colocalization, leaving its validity to be confirmed. This uncertainty was indeed the starting point and motivation for our study.

In our work, we simultaneously visualized flagellar filaments and core chemoreceptor proteins at the single-cell level in *P. aeruginosa*. We characterized the assembly and spatial coordination of the chemotaxis network and flagellar motor throughout the cell cycle, providing direct evidence of their colocalization and coordinated assembly. This represents a significant advance beyond prior indirect observations and supports the novelty of our study.

Accordingly, we have revised the relevant statements in lines 71-75 of the manuscript to better reflect the current state of the literature and emphasize the novelty of our direct observations.

*Line 108: "CheY has been shown to colocalize with chemoreceptors". The authors rely here (reference 29) and in other places on findings in E. coli. However, in the Introduction, they describe the many differences between the motility systems of P. aeruginosa and E. coli, e.g., the number of chemosensory systems and their spatial distribution (E. coli is a peritrichous bacterium, as opposed to the monotrichous bacterium P. aeruginosa). There seem to be proofs for colocalization of the Che and MCP proteins in P. aeruginosa, which should be cited here.*

Thank you for pointing this out. Harwood's group reported that a *cheY-YFP* fusion strain exhibited bright fluorescent spots at the cell pole, which disappeared upon knockout of *cheA* or *cheW*-genes encoding structural proteins of the chemotaxis complex. This strongly suggests colocalization of CheY with MCP proteins in *P. aeruginosa*. We have now cited this study as reference 17 in the manuscript.

*Figure 1B: Please replace the order of the schematic presentations, so that the cheY-egfp fusion, which is described first in the text, is at the top.*

We have modified the order of related images in Fig. 1B.

*Line 127: "by introducing cysteine mutations". Replace either by "by introducing cysteines" or by "by substituting several residues with cysteines".*

We changed the relevant statement to "by introducing cysteines".

*Line 144-145: "Given that the physiological and physical environments of both cell poles are nearly identical.". I think that also the physical, but certainly the physiological environment of the two poles is not identical. First, one is an old pole, and the other a new pole. Second, many proteins and RNAs were detected mainly or only in one of the poles of rod-shaped Gram-negative bacteria that are regarded as symmetrically dividing. Although my intuition is that the authors are correct in assuming that "it is unlikely that the unipolar distribution of the chemoreceptor array can be attributed to passive regulatory factors", relating it to the (false) identity between the poles is incorrect.*



We thank the reviewer for this important correction. We agree that the physiological environments of the two poles are not identical, given that one is the old pole and the other the new pole, and that many proteins and RNAs show polar localization in rod-shaped Gram-negative bacteria. Accordingly, we have revised the original text (lines 150-152) to read:

“Despite potential differences in the physical and especially physiological environments at the two cell poles, it is unlikely that the unipolar distribution of the chemotaxis complex can be attributed to passive regulatory factors.”

*Lines 151-154: "Considering the consistent colocalization pattern between chemosensory arrays and flagellar motors in P. aeruginosa". Does the word consistent relate to different reports on such colocalization or to the results in Figure 1D? In case it is the latter, then what is the word consistent based on? All together only 7 cells are presented in the 5 micrographs that compose Figure 1D (back to statistics...).*

We thank the reviewer for raising this point. To clarify, the word "consistent" refers to the observation of colocalization shown in Figure 1D & Figure S3. As noted in the revised figure legend for Figure 1D, a total of 145 cells with labeled flagella were analyzed, all exhibiting consistent colocalization between flagella and chemosensory arrays. Additionally, we have included a new image showing a large field of co-localization in the wild-type strain as Figure S3 to better illustrate this consistency.

*Figure 2A: Omit "Subcellular localization of" from the beginning of the caption.*

We removed the relevant expression from the caption.

**Reviewer #2 (Recommendations For The Authors):**

*I strongly recommend checking that CheY localizes to the receptor cluster in PA. This could be done by tagging cheA with a different fluorophore and demonstrating their colocalization. It would also be helpful to check that they are colocalized in the delta flhF mutant.*

We thank the reviewer for this valuable suggestion. We constructed a plasmid expressing CheA-CFP and introduced it into the CheY-EYFP strain by electroporation. Fluorescence imaging revealed a high degree of spatial overlap between CheA-CFP and CheY-EYFP signals, indicating that CheY-EYFP indeed marks the location of the chemoreceptor complex.

We have revised the manuscript accordingly (lines 118-123) and included these results in the new Fig. S2.

*The experiments under- and over-expressing CheY part seemed too unrelated to receptor-motor colocalization. I think the authors should think about a more direct way of testing whether colocalization of the motor and receptors is important for preventing signaling crosstalk. One way would be to measure cdG levels in WT and in delta flhF mutants and see if there is a significant difference.*

We thank the reviewer for raising the important point regarding potential cellular stress caused by elevated CheY concentrations, as well as for the suggestion to test the hypothesis using  $\Delta flhF$  mutants.

First, as noted in the response to your 2nd comment in Public Review, CheY-P concentration rapidly decreases away from the receptor complex. While deletion of *flhF* alters the position of the receptor complex, thereby shifting the region of high CheY-P concentration, it does not increase CheY-P levels elsewhere in the cell. Importantly, in the  $\Delta flhF$  strain, the receptor

complex and the motor still colocalize, so this mutant may not effectively test the role of receptor-motor colocalization in preventing crosstalk as suggested.

Regarding the possibility that elevated CheY levels stress the cells independently of CheY-P signaling, prior work in *E. coli* by Cluzel et al. (ref. 11) showed that overexpressing CheY several-fold did not cause phenotypic changes, indicating that simple CheY overexpression alone may not be generally stressful. Furthermore, our data indicate that the increase in c-di-GMP levels and subsequent cell aggregation upon CheY overexpression is not an all-or-none switch but occurs progressively as CheY concentration rises.

To further confirm that CheY overexpression promotes aggregation through increased c-di-GMP levels, we performed additional experiments co-overexpressing CheY and a phosphodiesterase (PDE) from *E. coli* to reduce intracellular c-di-GMP. These experiments showed that PDE expression mitigates cell aggregation caused by CheY overexpression (Fig. S8).

We have revised the manuscript accordingly (lines 290-294) and added these new results in Fig. S8.

**Reviewer #3 (Recommendations For The Authors):**

*(1) Can the authors elaborate more on the hierarchy of flagellar gene expression in P. aeruginosa and how this relates to their work?*

We thank the reviewer for the suggestion. We have now described the hierarchy of flagellar gene expression in *P. aeruginosa* in lines 341-348.

*(2) I would suggest that the authors check other flagellar mutants (than FliF and FliG) where the motor is partially assembled (e.g., any of the rod proteins or the P-ring protein), together with FlhF mutant, to see how a partially assembled motor affects the assembly of the chemosensory cluster.*

We thank the reviewer for this valuable suggestion. The P ring, primarily composed of FlgI, acts as a bushing for the peptidoglycan layer, and its absence leads to partial motor assembly. We constructed a  $\Delta flgI$  mutant and observed that the proportion of cells exhibiting distinct chemotactic complexes was similar to that of the wild-type strain, suggesting that the assembly of the receptor complex is likely influenced mainly by the C-ring and MS-ring structures rather than by the P ring. We have revised the original text accordingly (lines 217-220) and added the corresponding data as Figure S6.

*(3) I would suggest that the authors check the levels of CheY in cells induced with different concentrations of arabinose (i.e., using western blotting just like they did in Figure 3B).*

We have assessed the levels of CheY in cells induced with different concentrations of arabinose using western blotting, as suggested. The results have been incorporated into the manuscript (lines 274-275) and are presented in Figure S7.

*(4) To my eyes, most of the foci in FliF-FlhF mutant in Figure 3A are located at the pole (which is unlike the FlhF mutant in Figure 2). Is this correct? I would suggest that the authors also investigate this to see where the chemosensory cluster is located.*

We thank the reviewer for pointing this out. The distribution of the chemotaxis complex in the  $\Delta flhF \Delta fliF$  strain was investigated and showed in Fig. S4. Indeed, most of the chemoreceptor foci in this mutant are located at the pole. This probably suggests that, in the

absence of both FlhF and an assembled motor, the position of the receptor complex may be largely influenced by passive factors such as membrane curvature. This interesting possibility warrants further investigation in future studies.

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